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# Financial literacy pays: Smarter investing goes beyond AI

30 July 2026

Allianz Research

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Allianz School for Life

# Executive Summary


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- **Financial literacy is an important foundation for building, protecting and growing wealth over a lifetime, but it remains the exception.** From children's investment accounts in the US to pension reforms in Germany, responsibility for building wealth and financing retirement is increasingly shifting from institutions to individuals. As financial products become more complex and private provision more important, understanding basic financial concepts such as compound interest, inflation, risk and diversification is more important than ever. Yet financial capability is failing to keep pace. The fourth edition of the Allianz Research financial literacy study, questioning more than 8,000 respondents across eight countries, shows little change since our last survey in 2023[1]. Literacy remains worryingly low: only 17% scored as highly financially literate, while one in four scored low. The UK (23%), Austria (22%) and Germany (22%) record the highest share of highly financially literate respondents, Spain (17%), Italy (17%) and Poland (15%) occupy the middle ground, while the US (13%) and France (11%) rank at the bottom. The US also records the largest low-literacy group (33%), suggesting that greater capital-market participation does not automatically translate into greater financial capability.

- **Women, younger respondents and those less educated face the greatest financial challenges, but score lowest on financial literacy – a pattern that is remarkably consistent across countries.** Women need to finance longer retirements and accumulate private wealth, yet are only half as likely as men to achieve high literacy (11% vs. 24%). Younger generations also bear greater responsibility for privately funding their retirement as public pension systems come under growing pressure, yet they record the weakest results[2]: only 12% of Gen Z score as highly literate, compared with 22% of Baby Boomers. At the intersection of both divides, Gen Z women perform worst (10% high; 40% low), while Boomer men perform best (29% high; 13% low). Education is an even stronger divider: just 5% of respondents with primary education achieve high literacy, compared with 24% of those with tertiary education. Confidence and competence are also misaligned: While 17% of respondents score as highly literate, only 8% consider themselves more knowledgeable than the average investor. Conversely, the number of respondents who judge their financial knowledge as lower than average (48%) is double the number actually scoring as low-literate (26%). This presents the risk of too little confidence leading to inaction, while overconfidence may invite costly mistakes.

1 For the previous survey, see: Allianz Research (2023), "Playing with a Squared Ball: The Financial Literacy Gender Gap," https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/financial-literacy.html

2 See: Allianz Research (2026), "Pension Reform Survey 2026: Everyone knows reform is needed, few expect it to happen", https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/260625-pension-survey.html

- **Smarter investing, not higher saving, is one of the largest untapped sources of household wealth.** The return on savings depends not only on how much households set aside, but also on how effectively they invest. Financial literacy strongly shapes those choices: more than half of low-literacy respondents (53%) cannot identify an investment product they would prioritize, compared with only one in five (21%) of the high-literacy group. A simple simulation shows that shifting half of existing bank deposits in an average household portfolio, in equal parts, into national bonds and equities would have increased annual returns in every country, led by Poland, Germany, and Austria (each +1pp). Such a shift, without considering additional savings, would have lifted per capita financial wealth by 14–21% in real terms over the past two decades in countries with the highest deposit shares. Germany provides a clear example of the scale of the opportunity: the estimated per capita financial surplus over 20 years amounts to EUR11,700 after inflation.

- **AI is lowering the barriers to financial advice without raising capability, making financial literacy more important than ever.** AI makes financial advice more accessible by reducing the cost and increasing availability, but it cannot substitute for the financial literacy needed to evaluate that advice and make better money management decisions. AI adoption is already widespread: almost half of respondents (48%) use AI at least weekly, rising to 77% among Gen Z, while 14% already cite it as a main source of financial advice (26% among Gen Z). Yet greater access does not automatically translate into better decisions. AI users are considerably more confident in their financial knowledge (43% vs. 33% overall), but are no more likely to be highly financially literate (18% vs. 17%). This creates the risk of an overconfidence bias: AI may increase confidence without improving judgment. For financially literate households, AI can complement professional advice. For less financially literate households, however, it may reinforce poor decisions by providing advice they lack the knowledge to assess.

- **Closing financial literacy gaps requires a new policy playbook involving governments, employers and financial institutions.** There is an ever-growing need for credible financial and risk education. While the responsibility for financial security does not sit with households alone, people are being asked to take greater responsibility for their financial future and need the tools and knowledge to make informed decisions. Governments should embed financial education in schools and adult learning, while making the long-term consequences of financial decisions more tangible through pension dashboards and personalized retirement projections. Employers can strengthen financial capability through workplace education, pension guidance and access to trusted advice. Financial institutions should complement these efforts with simpler products, higher advice standards, digital tools and AI-enabled guidance that helps households move from saving to long-term investing. Across all stakeholders, interventions should be broad enough to reach everyone, yet sufficiently targeted to close the persistent literacy gaps among women, younger generations and less-educated adults. At the same time, interventions need to help people calibrate their financial confidence, recognize when professional advice is required and use AI as a complement, rather than a substitute, for trusted financial guidance.

# The financial literacy scoreboard: failing to reach the top

**In many advanced economies, responsibility for financial security is increasingly shifting from institutions onto individuals, raising the importance of financial literacy.** As social security systems come under increasing demographic strain, individuals need to take greater responsibility for securing their retirement through private plans[3]. At the same time, returns to capital continue to outpace those to labor – a development that could be turbo-charged by the impact of artificial intelligence on employment – making investments rather than wages the key driver of long-term wealth, whether the goal is homeownership or other financial objectives[4]. In this context, financial literacy equips people to make the decisions that build, protect and grow their wealth over a lifetime. The implications extend far beyond personal finances. Financial security underpins health, reduces stress and strengthens people's sense of control over the future[5]. As households assume greater responsibility for retirement planning and wealth accumulation, financial literacy increasingly determines not only who prospers financially, but who enjoys greater economic resilience and wellbeing. And the challenge is only becoming harder. A rapidly evolving financial landscape – defined by digital assets, increasingly sophisticated financial products and persistent cost-of-living pressures – demands ever greater confidence and competence in managing money.

**To understand how well people are equipped to navigate this new financial reality, we surveyed more than 8,000 respondents across eight countries: Austria, France, Germany, Italy, Poland, Spain, the UK and the US[6]**. The survey, the fourth edition of our financial literacy study, builds on Allianz Research's financial literacy series, using the same core questionnaire to track developments over time[7]. This year's edition expands the country coverage with the addition of Austria and Poland, with the latter offering a first perspective on financial literacy in Central and Eastern Europe. The

3 Recent pension reform proposals increasingly emphasize funded private retirement savings. In Germany, these include plans for a capital-funded private pension pillar and the introduction of a "Frühstart-Rente", a state-supported investment account for children, designed to encourage long-term capital-market participation from an early age.

4 See Jordà, O., K. Knoll, D. Kuvshinov, M. Schularick and A. Taylor (2019), "The Rate of Return on Everything, 1870–2015," Quarterly Journal of Economics, 134(3), 1225–1298. Using long-run historical data for 16 advanced economies, the authors find that the rate of return on wealth has persistently exceeded the rate of economic growth over the past 150 years, with average returns on capital roughly twice as high as economic growth.

5 See: OECD (2023), "International Survey of Adult Financial Literacy," OECD Publishing, Paris. The survey finds that adults with high financial literacy report higher financial well-being and financial resilience than those with low literacy, and that financial literacy helps individuals "develop greater control over personal financial matters."

6 Details on the sample are provided in Appendix A. The complete questionnaire is reproduced in Appendix B.

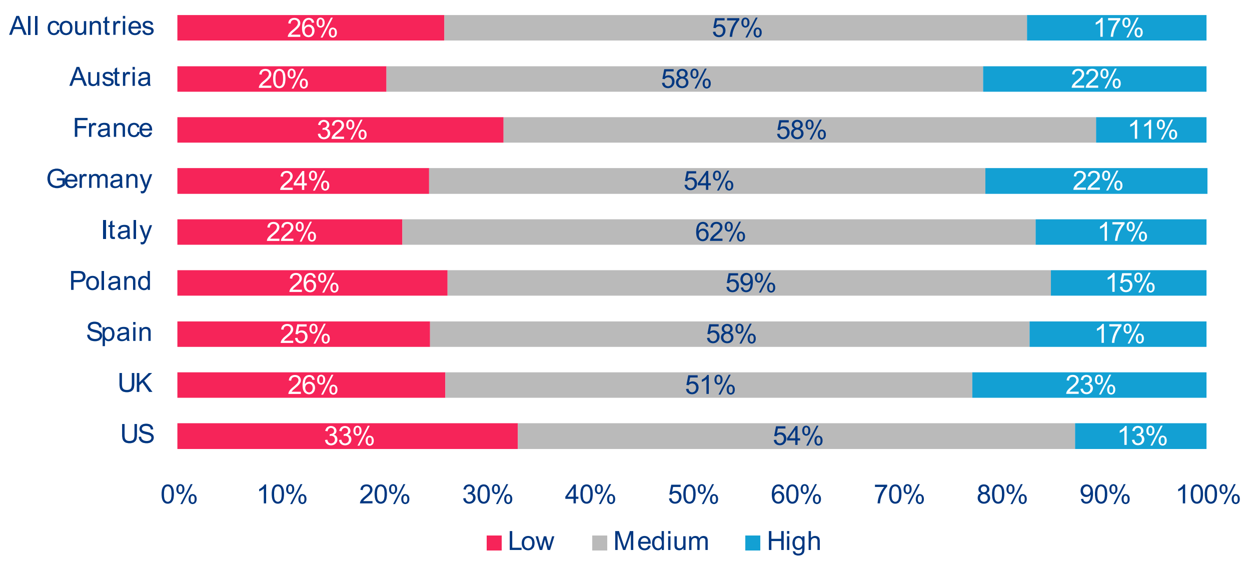
7 To ensure comparability across countries and survey waves, we measure financial literacy using the same index in each survey wave. For the previous survey, see: Allianz Research (2023), "Playing with a Squared Ball: The Financial Literacy Gender Gap," https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/financial-literacy.html

financial literary score is based on nine questions covering the core concepts underpinning sound financial decision-making – interest, compound interest, inflation, risk diversification and probability – drawing on the seminal work of Professors Annamaria Lusardi and Olivia Mitchell[8]. Respondents are grouped into three levels, or bands, of financial literacy according to the number of correct answers: low (0–2), medium (3–6) and high (7–9). This classification is used throughout the report.

**The survey confirms an insufficient level of financial literacy almost everywhere.** Only 17.5% of all respondents reach the high-literacy band, while 25.9% fall into the low band. The majority of 56.6% are classified as medium, suggesting a basic grasp of financial concepts, but not the level of knowledge required to navigate increasingly complex financial decisions. Notably, this medium share is strikingly uniform across countries, hovering around 55% almost everywhere, while the differences between countries play out almost entirely in the relative size of their high- and low-literacy bands (Figure 1). Ranked by the shares of respondents with high levels of financial literacy, the UK (22.9% high literacy), Austria (22.0%) and Germany (21.5%) record the strongest results. Spain (17.4%), Italy (16.9%), and Poland (15.4%) occupy the middle ground, while France (10.7%) and the US (12.9%) rank at the bottom. Among these country differences, the US stands out in particular. Despite having one of the world's most market-based retirement systems, it also records the largest low-literacy group (33.0%). This may be no coincidence. Through 401(k)s, IRAs and default investment options, households can participate in capital markets with little active financial knowledge: contributions are automatic, funds are pre-selected and investing happens largely on autopilot. Being invested, therefore, does not necessarily translate into greater financial capability.

**The financial literacy scoreboard has moved little since 2023, although some countries have made limited progress.** Among the six countries included in both the 2023 and 2026 surveys, Germany and the UK record the notable improvements, with the share of respondents in the high-literacy band increasing by 5.4 and 7.0pp, respectively. The US also sees a moderate improvement in the share of respondents with high literacy (+2.6pp), while France records only marginal progress (+1.0pp). Spain sees a growing polarization of financial literacy: both the low- (+4.4pp) and high-literacy groups (+1.1pp) grow, while the medium-literacy

**Figure 1:** Financial literacy levels, share of respondents by country in %




Source: Allianz Research

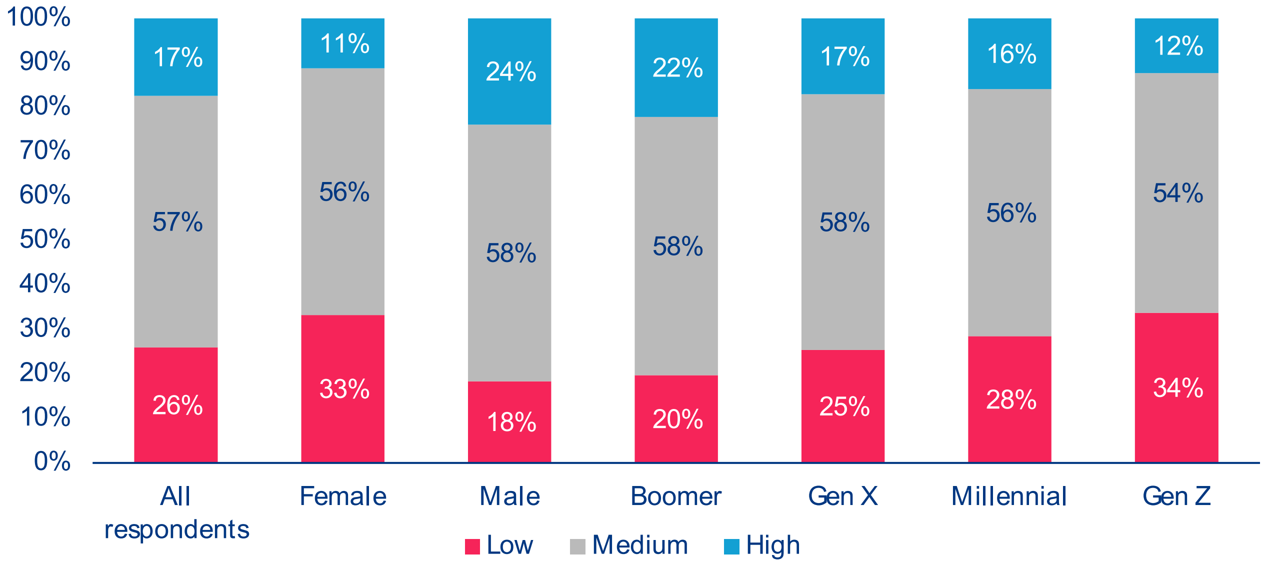
8 The index follows the « Big Three » and extended financial-literacy questions developed in Lusardi, A. and Mitchell, O. S. (2011), «Financial Literacy around the World: An Overview», Journal of Pension Economics and Finance, 10(4), 497–508, and Lusardi, A. and Mitchell, O. S. (2014), «The Economic Importance of Financial Literacy: Theory and Evidence», Journal of Economic Literature, 52(1), 5–44.

group shrinks (-5.5pp). In contrast, Italy remains broadly unchanged, showing neither major improvement nor deterioration. Overall, the findings suggest that current financial literacy initiatives are failing to deliver broad-based gains, underscoring the need for more sustained and better-targeted interventions.

**Male and older players know the game: financial literacy splits sharply by gender and rises steadily with age (Figure 2).** First, the gender literacy gap is striking: 23.9% of men reach the high-literacy band, compared with just 11.2% of women. One-third of women (33.3%) fall into the low-literacy band against fewer than one in five men (18.4%). This gap is particularly concerning because women typically live longer, earn less over their lifetime and therefore depend even more on making their savings work effectively. Second, financial literacy increases across the generations[9]: only 12.3% of Gen Z reach the high-literacy band, compared with 22.2% of Baby Boomers, while the low-literacy share falls from 33.8% to 19.7%. This is a worrying mismatch. Younger generations will need to start building private wealth earlier and more consistently than previous cohorts, as they are likely to depend more heavily on defined-contribution pensions and their own savings. Yet they are the least financially equipped to navigate these responsibilities. Together, these two divides reinforce one another: younger generations and women face the greatest challenge in securing their financial future and record the weakest financial literacy.

**A closer look shows that the gender gap is remarkably persistent across countries. In every country, men are around twice as likely as women to reach the high-literacy band (Figure 3).** The gap is largest in Austria (30.4% vs 13.5%), Germany (29.5% vs 13.5%) and the UK (30.5% vs 15.6%), meaning that these countries' strong overall performance is driven primarily by men. This matters because women typically accumulate less income and wealth over their lifetime while depending more heavily on it in retirement[10]. Therefore, each euro of women's savings and investments matters even more. Yet they are also worse served when they seek financial advice[11]. Lower financial literacy therefore compounds existing gender disparities rather than offsetting them.

**Figure 2:** Financial literacy levels, share of respondents by gender and generation in %



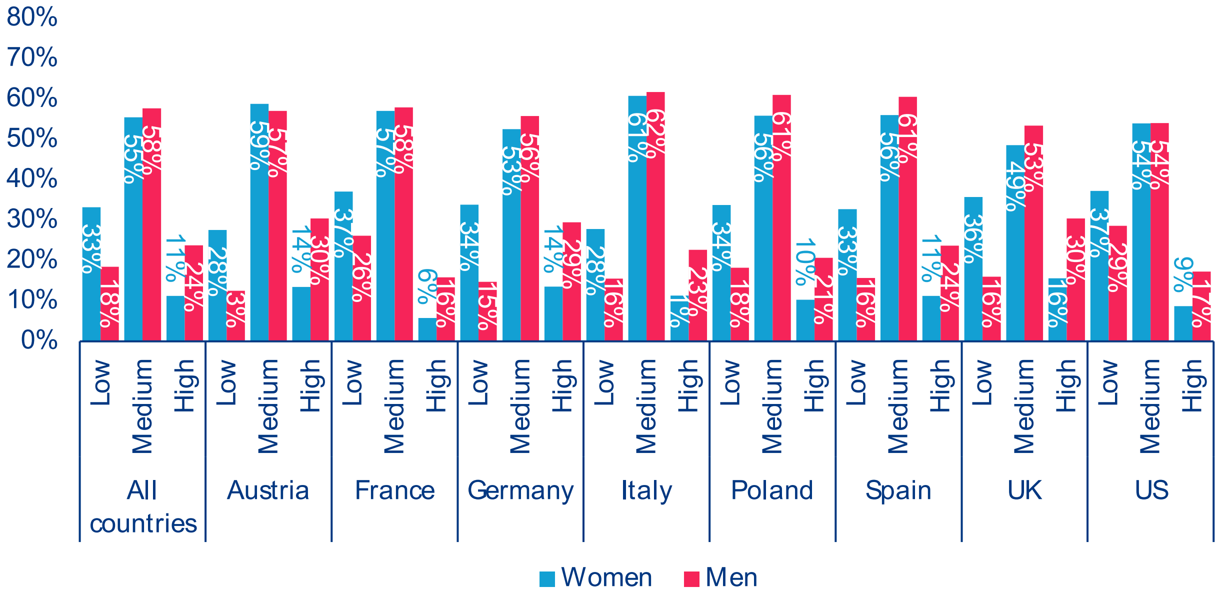

Note: Baby Boomers (birth years 1946–1964), Generation X (1965–1980), Millennials (1981–1996) and Generation Z (1997–2012).
Source: Allianz Research

9 Generations are defined according to the Pew Research Center: Baby Boomers (birth years 1946–1964), Generation X (1965–1980), Millennials (1981–1996) and Generation Z (1997–2012).

10 See Allianz Research (2026), "Closing the Gender Income Gap: From Paycheck to Pension," https://www.allianz.com/en/economic_research/insights/publications/specials_fmo/260305-gender-gap.html. The study estimates a lifetime income gender gap of around 16% for the 2000 birth cohort across 14 OECD countries, driven primarily by labor income, and finds that strengthening financial literacy can raise women's annual investment returns by up to 1.5pp. On pensions specifically, Eurostat (2026) reports that women aged 65+ in the EU received on average 24.5% less pension income than men in 2024.

11 See: Bucher-Koenen, T., Hackethal, A., Koenen, J. and Laudenbach, C. (2025), "Gender Differences in Financial Advice," American Economic Review, 115(12), 4218–4252. Drawing on some 27,000 real-world advisory meetings at a large German bank, the authors find advisors give more self-serving advice to women, recommending expensive in-house products more often and offering fee rebates less often.

**Figure 3:** : Gender gap in financial literacy, share of female and male respondents by country in %

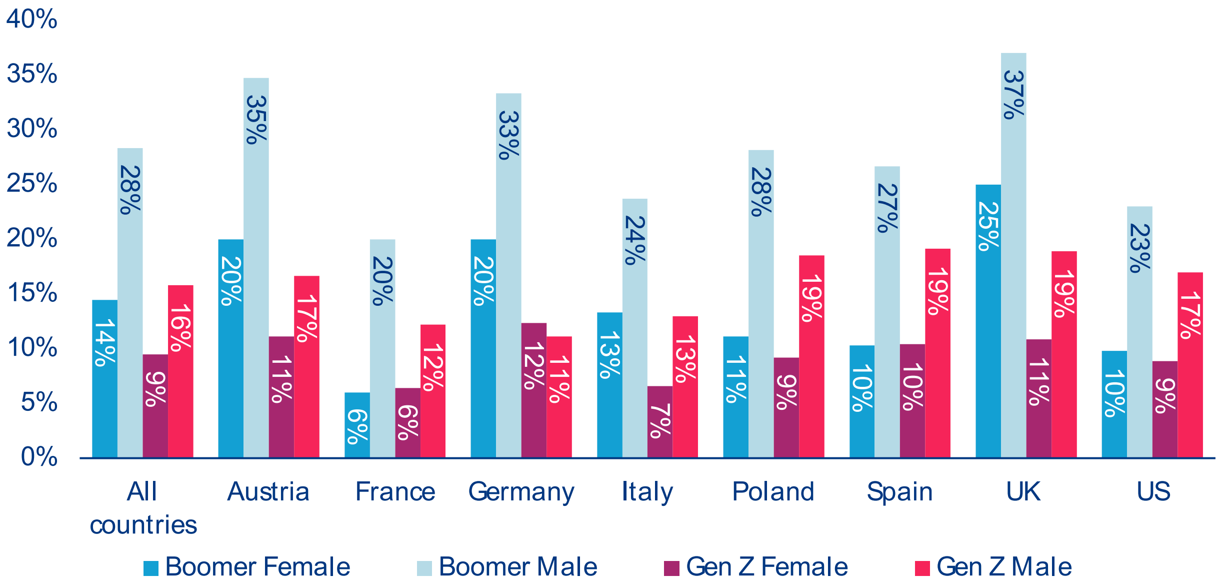



Source: Allianz Research

**The gender and generational divides reinforce each other. Gen Z women record the weakest results of any demographic group,** with only 9.6% reaching the high-literacy band and 39.8% falling into the low band. In comparison, Baby Boomer men perform best, with 28.9% in the high-literacy band and just 13.2% in the low band (Figure 4). This pattern is particularly pronounced in Italy, where 23.7% of male Baby Boomers are highly financially literate but only 6.6% of female Gen Z – nearly a fourfold difference. The US is a notable exception, with financial literacy remaining comparatively flat across generations, echoing the earlier finding that widespread market participation does not necessarily translate into greater financial capability. Overall, the evidence points in one direction: the groups carrying the greatest responsibility for financing their own future are also the least financially prepared.

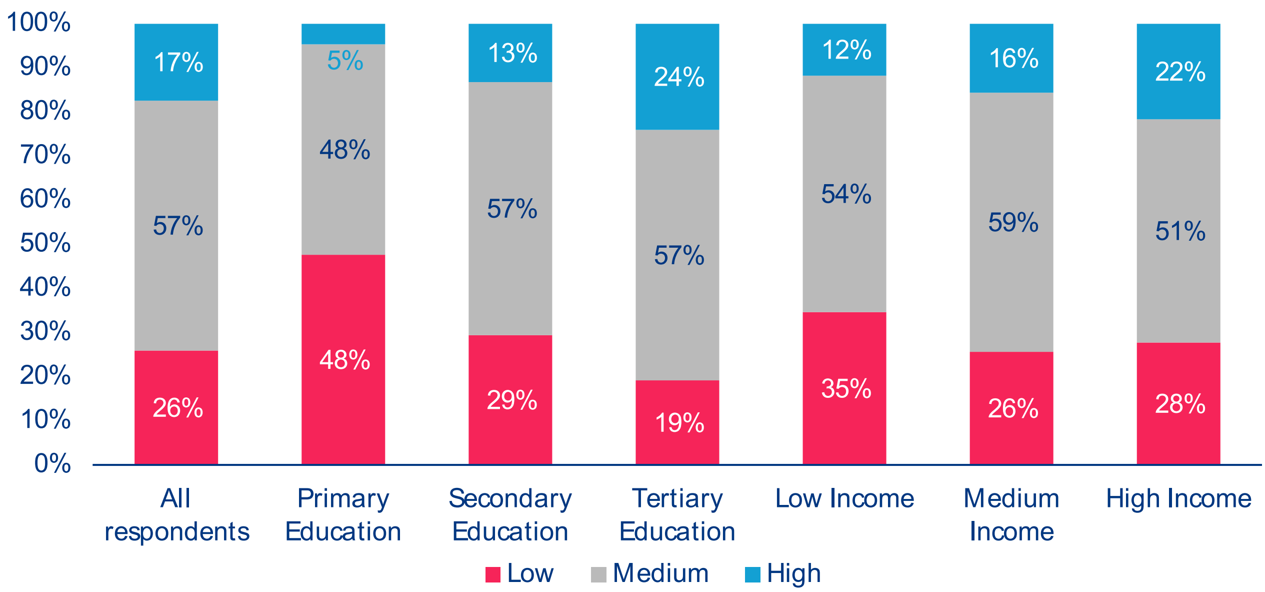
**Education and income sort the remaining field, and financial literacy risks entrenching the divides they reflect.** The gradients here are the steepest in the entire survey (Figure 5). Among respondents with only primary education, 47.8% fall into the low-literacy band and just 4.7% reach the high band. By contrast, 24.0% of respondents with tertiary education reach the high-literacy band – more than five times the share among the primary educated. Income shows a similar, though weaker, gradient: 21.7% of high-income respondents reach the high-literacy band, compared with 11.8% of the low-income group. The implication is that financial literacy gaps may reinforce existing economic inequalities.

**Figure 4**: High financial literacy generational and gender divide, share of respondents in the high literacy band by country, gender and generation in %




Source: Allianz Research

**Figure 5:** Financial literacy levels, share of respondents by education and income in %

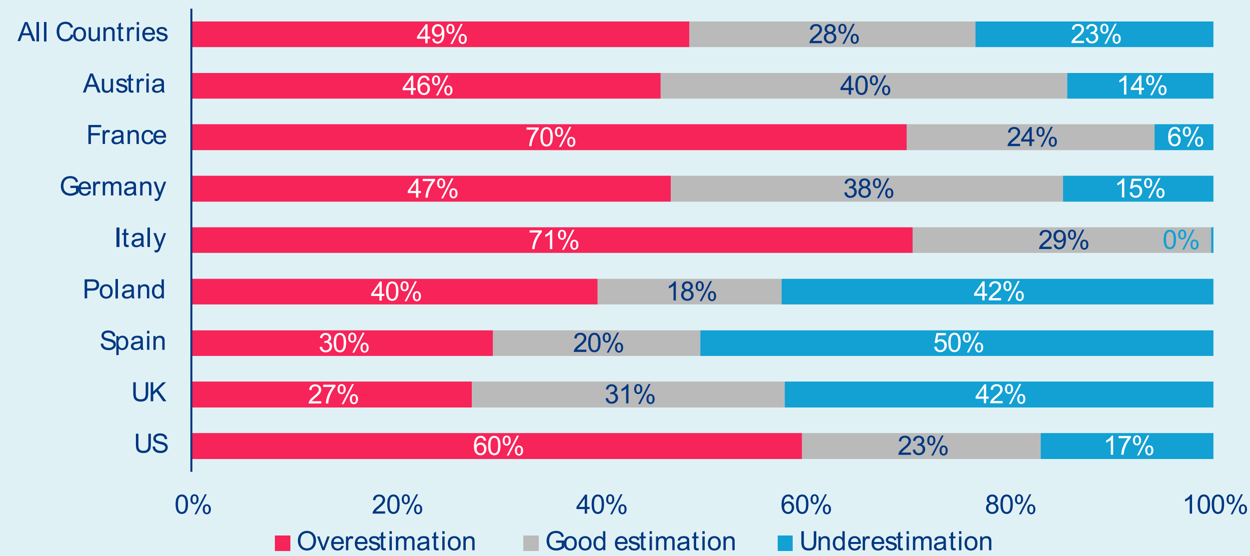



Source: Allianz Research

## Box 1: Can people read the price tag? Inflation perceptions and the cost-of-living barrier

**Financial literacy is not only about understanding financial concepts but also about interpreting the economic signals that shape everyday decisions, with inflation being one of the most important.** Households that overestimate inflation underestimate the real return on their savings and may conclude they have too little left to invest, reinforcing the already existing cost-of-living pressures. We therefore asked respondents to estimate the current inflation rate in their country, counting answers within one percentage point of the official rate as accurate (Figure 6). The results are sobering: only 28.0% estimated inflation correctly, while 48.7% overestimated it and 23.3% underestimated it. On average, respondents put inflation at 5.7% compared with the actual rate of 2.9%. Overestimation is particularly widespread in Italy (70.6%), France (70.0%) and the US (59.7%). Underestimation is most common in Spain (50.2%) and the UK (41.9%). Both distort financial decisions, but persistent overestimation is likely to discourage investing by leading households to underestimate the purchasing power of future investment returns.

**Figure 6:** Inflation perceptions, share of respondents by country underestimating, accurately estimating, and overestimating current inflation, in %

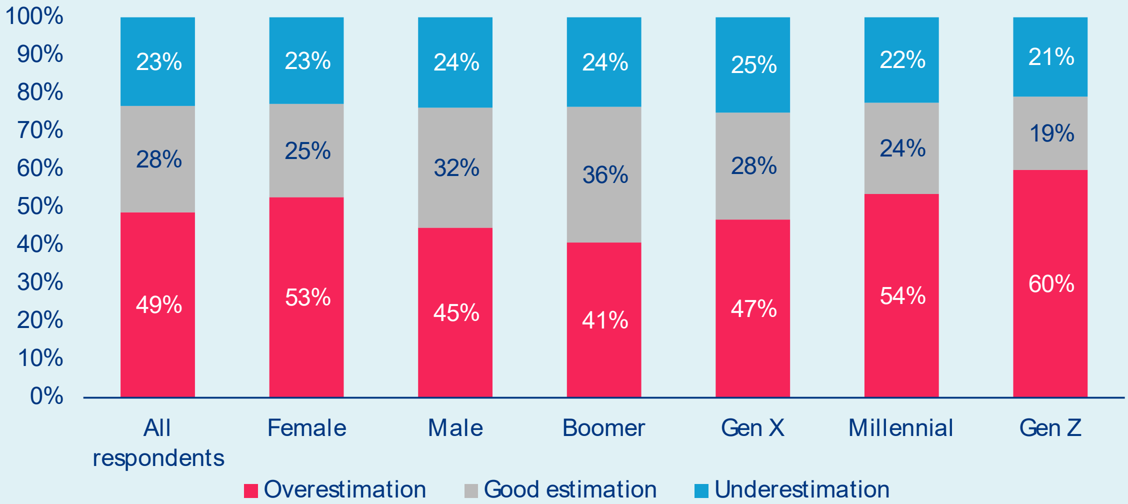



Notes: We consider estimates within a 1pp range around current inflation as a good estimate.

Sources: LSEG Datastream, Allianz Research

**Inflation perceptions closely mirror the fault lines of financial literacy scoreboard.** Austria (39.8% accurate) and Germany (38.4%) record the highest shares of correct estimates, while countries with weaker financial literacy also display larger misperceptions. The same pattern appears across demographic groups (Figure 7): men estimate inflation more accurately than women (31.6% vs 24.6%), and accuracy declines steadily from Baby Boomers (35.7%) to Gen Z (19.3%), who are also the most likely to overestimate inflation (59.9%). Reading the price tag, in other words, is another expression of financial literacy: the same groups that struggle most with financial concepts are also the least able to accurately assess the economic environment in which they make financial decisions.

**Figure 7:** Inflation perceptions, share of respondents by gender and generation underestimating, accurately estimating, and overestimating current inflation, in %



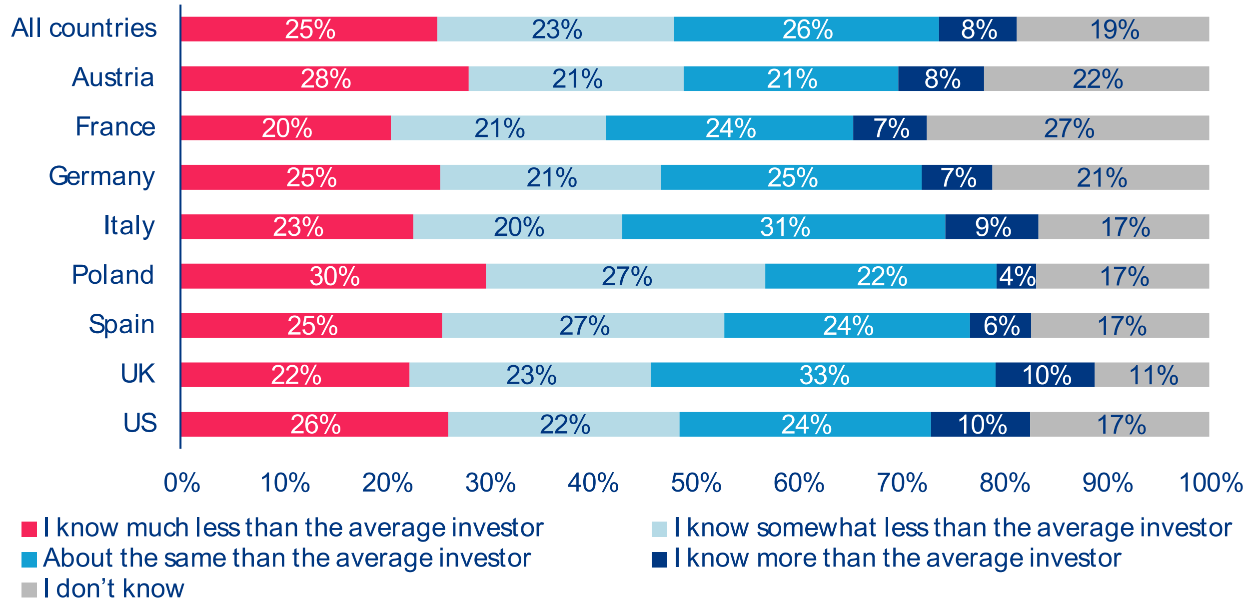

Notes: We consider estimates within a 1pp range around current inflation as a good estimate.

Sources: LSEG Datastream, Allianz Research.

# Financial confidence does not always follow financial competence

**Financial knowledge is only part of the equation; people must also feel confident enough to act on it.** Asked to compare their financial knowledge with that of the average investor, respondents paint a remarkably cautious picture. Across all eight countries, one in four (25.0%) believe they know much less than the average investor, while a further 23.0% consider themselves somewhat less knowledgeable (Figure 8). Just over one-quarter (25.8%) rate themselves as about average, and only 7.6% believe they know more than the average investor. Nearly one in five respondents (18.7%) are unable to judge their own level of knowledge. Country differences are relatively modest. Poland records the lowest level of self-confidence, with 56.9% placing themselves below the average investor and only 3.9% above, while respondents in the UK and the US express the greatest confidence, with 9.6% considering themselves more knowledgeable than average.

**Figure 8:** Financial confidence, share of respondents by country in %



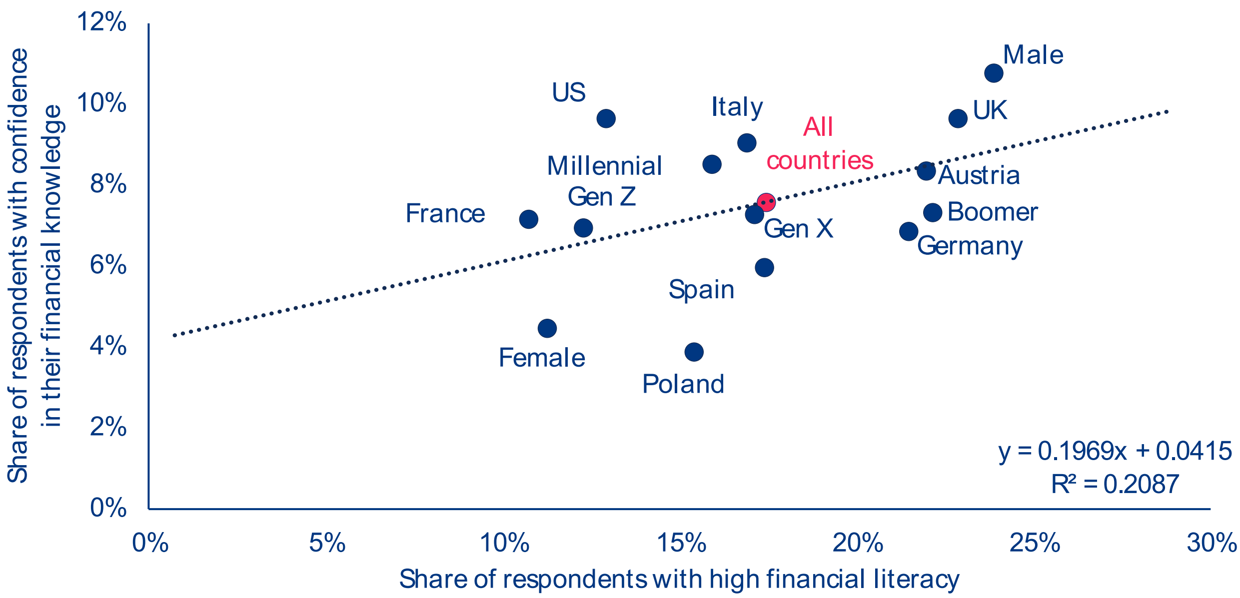

Source: Allianz Research

**Confidence is not always an accurate reflection of competence.** Across all eight countries, 17.5% of respondents reach the high-literacy band, but only 7.6% consider themselves more knowledgeable than the average investor. Conversely, almost half of respondents (48.0%) judge their financial knowledge to be below average, even though only 25.9% belong to the low-literacy band. Confidence does rise with financial literacy – the share considering themselves more knowledgeable than average increases from 4.6% in the low-literacy band to 6.0% in the medium band and 16.9% in the high band – but much more slowly than knowledge itself. The relationship also varies across countries (Figure 9). The UK and the US are the most self-assured (9.6% highly confident in each), yet while the UK combines this with one of the highest levels of financial literacy, the US records the lowest share of highly financially literate respondents (12.9%), making it the most overconfident country in the sample. Poland presents the opposite pattern: only 3.9% consider themselves above average despite a comparatively larger high-literacy group (15.4%), suggesting confidence there falls short of competence.

**The gender divide is stronger than the generational divide: women's confidence is even lower than their financial literacy.** While men are overconfident relative to their knowledge – 23.9% are highly literate, but only 10.8% are highly confident – women show the opposite and more damaging pattern. They pair an 11.3% highly literate share with just 4.5% high confidence, which are two of the widest gaps in the dataset (12.6 percentage points in literacy and 6.3 percentage points in confidence). These gaps reinforce each other and risk keeping capable women out of financial markets they could navigate perfectly well. Generations drift apart the same way: literacy climbs with age but confidence stays flat, so Baby Boomers top the literacy table (22.2%) yet claim only 7.3% high confidence, while Millennials (15.9% vs 8.5%) and Gen Z (12.3% vs 6.9%) tilt toward overconfidence. Experience, it seems, teaches humility as readily as it builds knowledge.

**The mismatch between knowledge and confidence matters for financial outcomes.** Overconfidence, or when confidence exceeds actual knowledge, invites costly mistakes, as people overestimate their ability to judge risk. Knowledge without confidence is just as limiting, keeping capable savers on the sidelines, postponing decisions or leaning too heavily on others. The goal of financial education should therefore be broader than raising test scores. It should help people calibrate their confidence – to trust what they know, recognize what they do not, and seek reliable advice when needed.

**Figure 9:** Financial literacy versus financial confidence, share of respondents by country, gender and generation in %

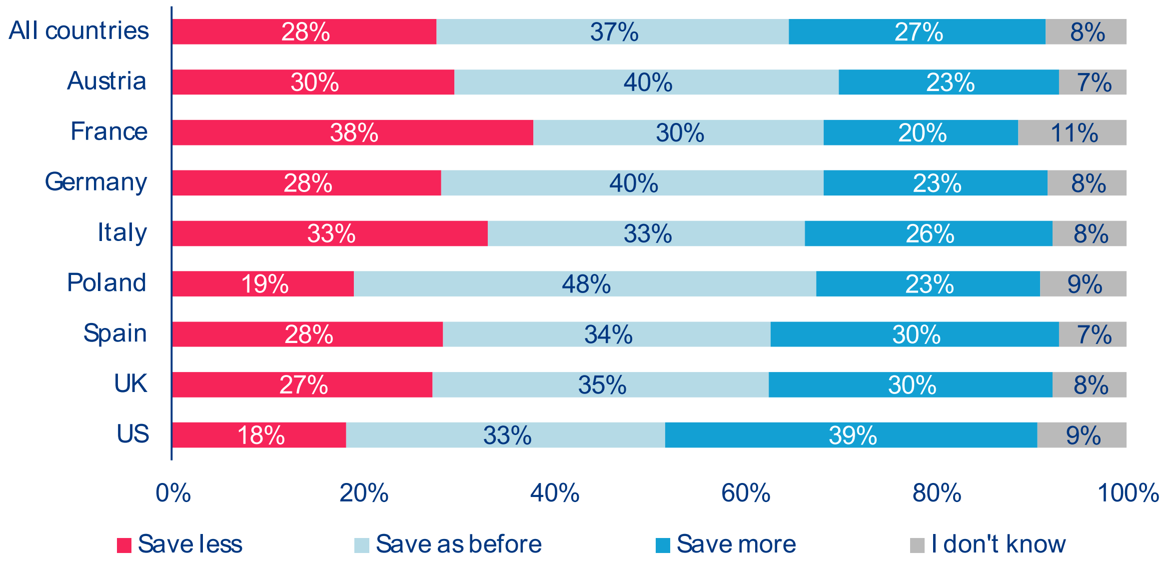



Source: Allianz Research

# Saving intentions are stable, investment choices make the difference

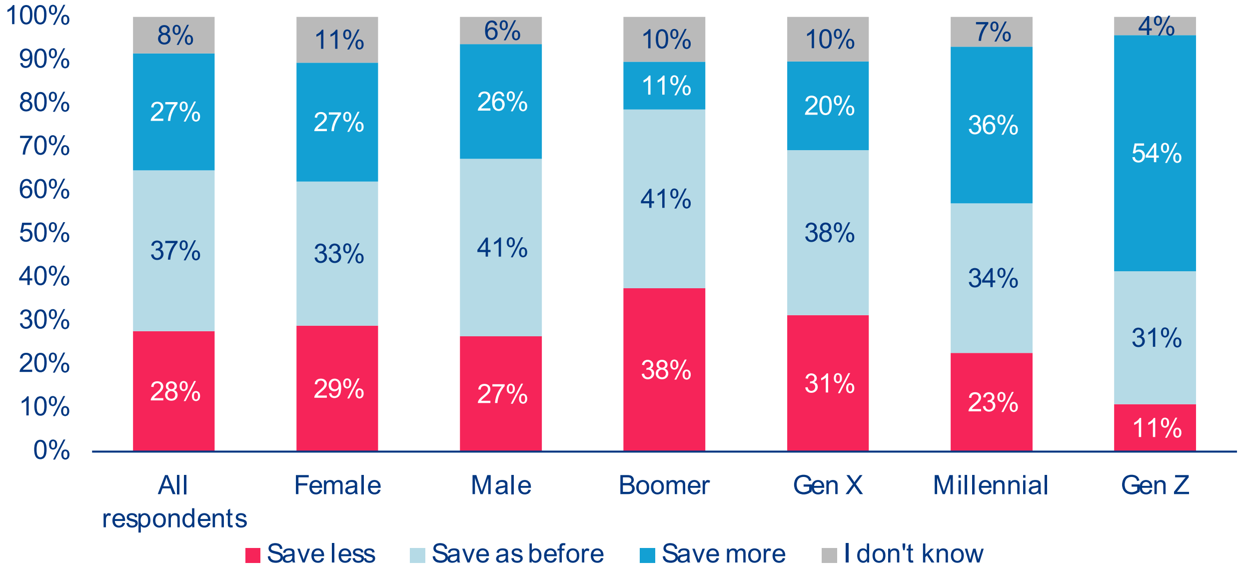
**Building wealth depends on two decisions: how much households save and how they invest those savings.** We therefore asked respondents about both their future saving intentions and the products they expect to give greater weight in their portfolios. While saving intentions also provide an indication of perceived financial constraints, they reveal remarkably little appetite for a broad shift in household saving. Across all countries, 37% expect to save as before, 27% plan to save more and 28% to save less, while 8% remain undecided (Figure 10). The pattern is broadly similar across countries, with France standing out as the only country where the largest group plans to save less (38%) and the US as the only one where the largest group intends to save more (39%). These differences partly reflect national saving patterns. Household saving rates remain above their pre-pandemic levels in every country except the US, leaving American households with the most ground to recover after the post-Covid decline.

**Figure 10:** Will your savings behavior change?, share of respondents by country in %




Source: Allianz Research

**Figure 11:** Will your savings behavior change?, share of respondents by country in %



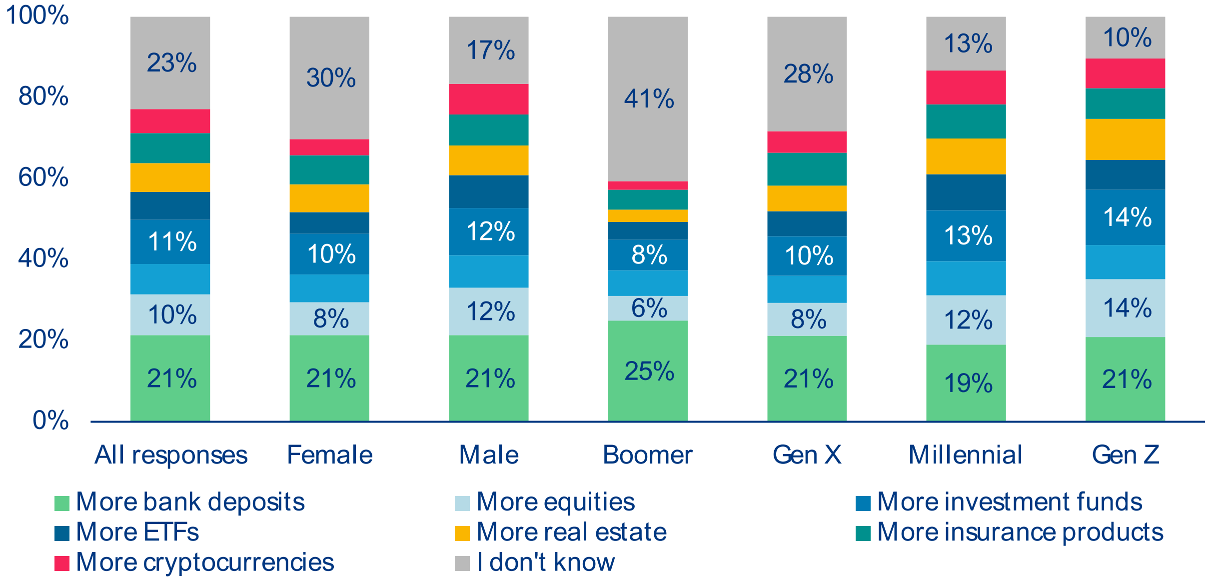

Source: Allianz Research

**A much clearer pattern emerges across generations (Figure 11).** More than half of Gen Z respondents (53%) and over one-third of Millennials (35%) intend to save more, compared with just 19% of Gen X and 11% of Baby Boomers. Conversely, the share planning to save less rises steadily with age, from 12% among Gen Z to 38% among Boomers, closely reflecting the life-cycle hypothesis and observed saving behavior. The broader message, however, is that most households do not expect to change how much they save. If long-term wealth creation is to improve, the greater opportunity lies not in saving more, but in investing existing savings more effectively.

**The survey suggests that investment choices are where the real differences begin.** When asked which products they intend to give greater weight in future, men allocate 40% of their choices to capital-market instruments – equities, bonds, investment funds and ETFs – compared with 30% among women (Figure 12). Bank deposits remain the single most popular choice for both groups (21%), while real estate and insurance attract similar shares. The largest difference is uncertainty: 30% of women answer „I don't know," almost twice the share among men (17%). The much higher share of undecided responses echoes the confidence gap identified earlier.

However, these results come with a caveat: They are stated intentions rather than actual portfolio allocations, but they nevertheless suggest that differences in wealth accumulation are likely to arise less from access to financial markets than from knowledge and confidence in navigating them.

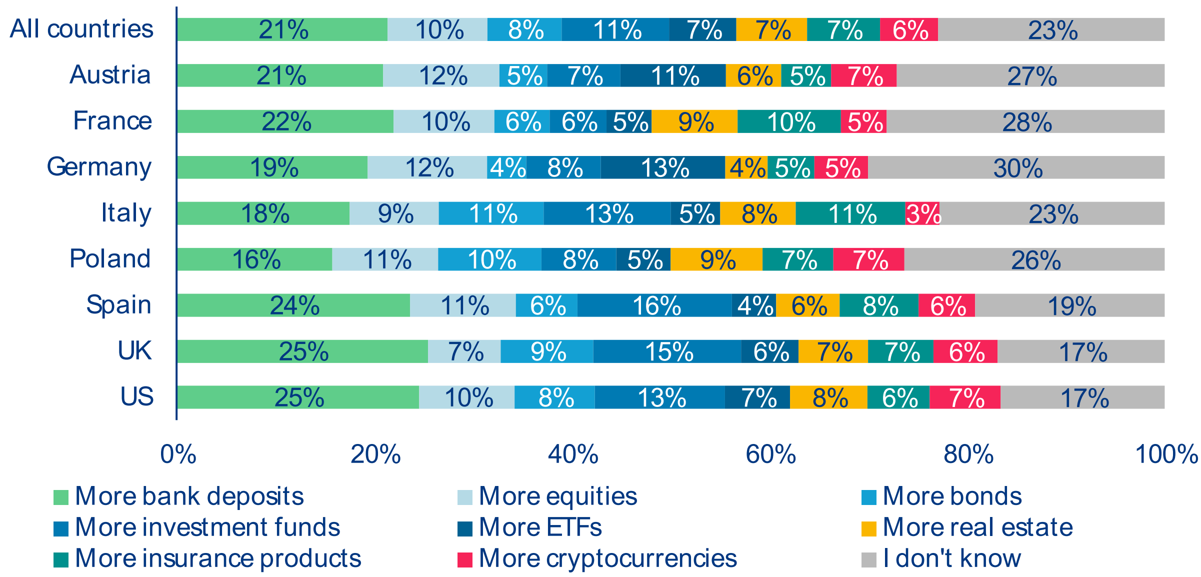
**Figure 12:** Which savings products will you give in future more weight in your portfolio?, share of responses by gender and generation in %




Source: Allianz Research

**Younger generations and male respondents have the highest appetite for capital-market investments.** Those products account for 44% of the intended allocations of Gen Z respondents and 42% of Millennials, compared with 30% for Gen X and just 24% for Baby Boomers. At the same time, indecision rises sharply with age, from only 10% among Gen Z to 41% among Boomers. Even among Baby Boomers, however, securities remain the preferred choice over deposits in most countries, with the UK and France the only exceptions. Millennials also show a stronger preference for real estate, consistent with being at the life stage when home ownership is still being built. The intersection of age and gender reinforces these differences (Figure 13). Older women are by far the most hesitant and least knowledgeable investors, with more than half answering „I don't know" in several countries. In Poland, as many as 64% gave this answer, although younger Polish women increasingly resemble their male peers, generally preferring securities over deposits and reporting much lower levels of uncertainty. Men remain more decisive across all generations. Taken together, the findings suggest that future differences in household wealth will depend less on how much people save than on how confidently and effectively they put those savings to work.

**Figure 13:** Which savings products will you give in future more weight in your portfolio?, share of responses by country in %

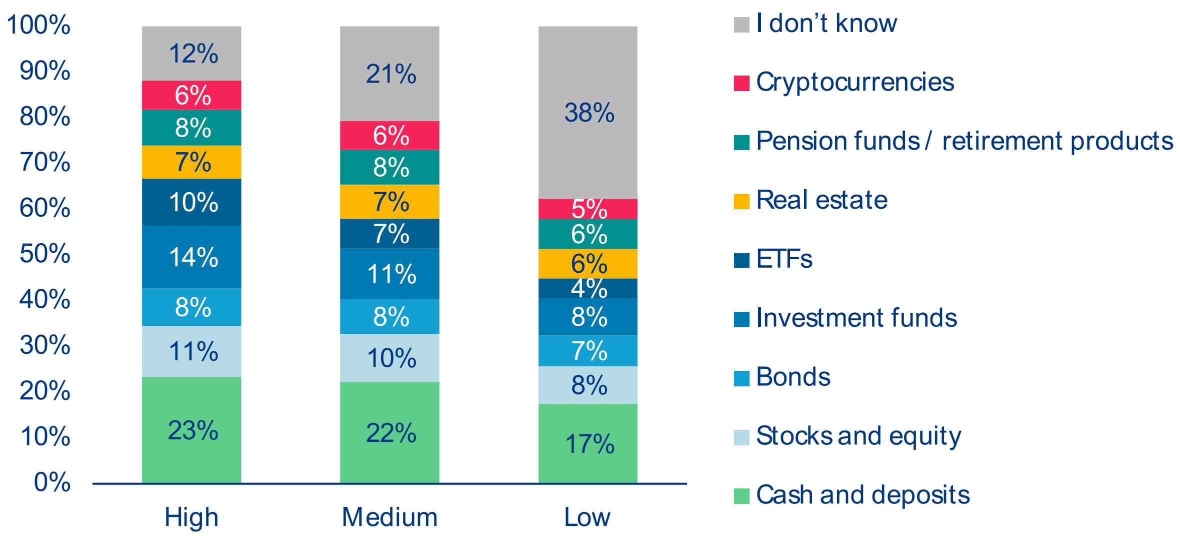



Source: Allianz Research


# The price of not knowing is high

**While many people know they should save more, a considerable share – cutting across both generational and gender lines – do not know where to put their money.** When asked which savings products they plan to give more weight to in the future, and sorting the answers by financial literacy – low, medium, and high – a clear and consistent pattern emerges: financial literacy plays a decisive role in shaping where people plan to allocate their savings. While traditional products remain broadly relevant across all groups, more sophisticated investment instruments are disproportionately favored by the financially literate (see Figure 14).

**Figure 14:** Which savings products will you give more weight to in your portfolio in the future?, share of responses by financial literacy level in %




Source: Allianz Research

**The more you know, the more you diversify.** The intention to increase exposure to equities, investment funds and particularly exchange-traded funds (ETFs) rises markedly with financial literacy. Less financially literate households show lower propensity to move into these asset classes. ETFs, long favored by cost-conscious and informed investors alike, attract nearly three times as much enthusiasm from high-literacy respondents as from low-literacy ones – reflecting not only greater familiarity with these instruments, but also a more strategic approach to long-term wealth building.

**Not everything divides along literacy lines.** Products such as real estate and pension or retirement-related savings display a much flatter distribution across literacy groups, driven less by financial knowledge and more by structural factors such as life-cycle considerations or institutional frameworks. Cryptocurrencies tell a similar story: the intention to increase exposure to crypto assets shows only limited variation across literacy levels, suggesting that participation in this segment reflects broader trends, accessibility or speculative motives rather than financial knowledge.

**Cash and deposits tell a surprising story.** Safe and liquid assets continue to feature prominently across all levels of financial literacy, suggesting that precautionary motives shape saving behavior across the board, regardless of financial sophistication. Contrary to what one might expect, it is the most financially literate respondents who express the greatest preference for cash and deposits (almost 41%), ahead of both the medium- and low-literacy groups (nearly 25% and 35%, respectively). This likely reflects not only precautionary savings but also a more deliberate portfolio management mindset – using liquid assets as a conscious strategic component rather than a default.

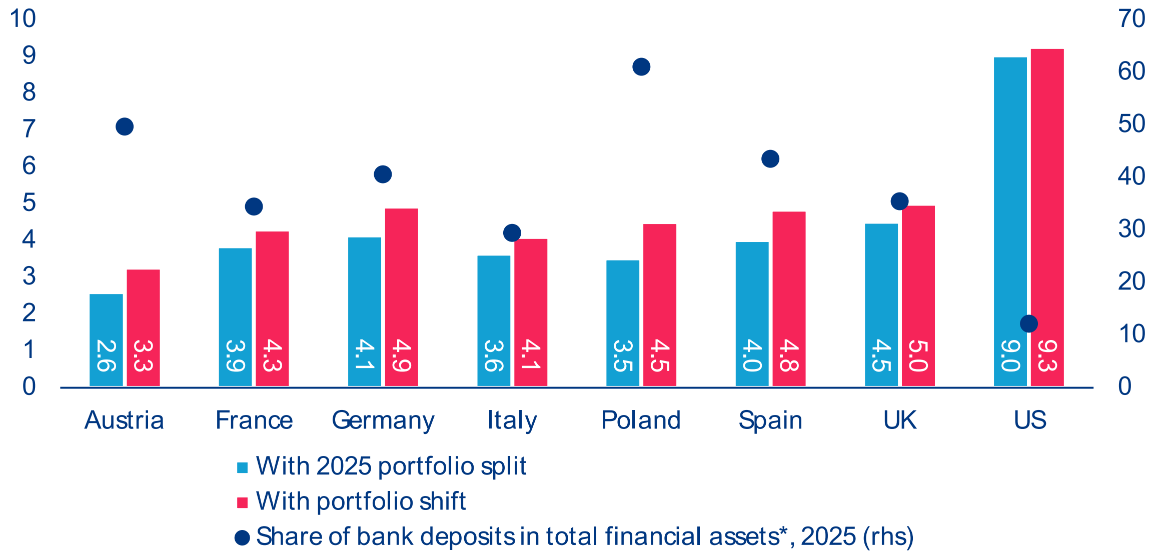
**Uncertainty is the real dividing line.** The starkest finding in the data is the share of respondents who answered "I don't know." Among low-literacy respondents, more than half (almost 53%) could not identify a savings product they would prioritize. This figure drops to roughly 33% for medium-literacy and just under 21% for high-literacy respondents. A lack of financial knowledge does not just influence what people invest in – it leaves many unable to make any informed choice at all. Rather than being put to work, savings remain parked in cash or low-interest bank deposits.

**Knowing matters – but how much exactly?** To put a number on it, we constructed a simple simulation based on the current average portfolio structure of households across the eight countries in our survey. If the average saver had invested their financial wealth[12] in 2005 according to their current portfolio allocation and held it for 20 years, what would their average annual return have been? We then make a key comparison: what if those same households had made one modest adjustment – redirecting half of their bank deposit holdings, in equal parts, into national bonds and equities? In every country, the answer points the same way: they would have done better. The return gap tracks closely with the share of bank deposits in total financial assets – a relationship that is as intuitive as it is consequential (see Figure 15). Poland, where deposits account for more than 60% of household financial wealth, sees the largest improvement: an actual nominal return of 3.5% per year would have risen to 4.5%. Germany and Austria, both carrying deposit shares well above 40%, would have gained 0.8 and 0.7pp respectively. The United States, where deposits represent only around 12% of financial assets and equity ownership is underpinned by the widespread use of employer-sponsored retirement plans such as the 401(k), starts from a much higher base – 9.0% per year – and has correspondingly little to gain from rebalancing. The pattern is consistent: the more a household relies on deposits, the more return it leaves behind.

**Stretched over two decades, modest annual differences add up to significant wealth gains.** These additional returns – and the financial surplus they generate over time – require no additional saving from current income: they arise solely from reallocating existing wealth from low- to higher-yielding assets. Countries with the highest deposit shares would see double-digit wealth gains adjusted for inflation: on average around 14% in Austria, 16% in Germany, and over 21% in Poland. In per capita terms, the nominal financial surplus, or unadjusted cash savings, ranges from an average of nearly EUR1,600 in Poland, where financial wealth is still comparatively small, to almost EUR10,300 in Austria and just under EUR17,600 in Germany (see Figure 16). Once inflation is stripped out, the range is from around EUR800 in Poland to nearly EUR6,100 in Austria and around EUR11,700 in Germany. A word about the US households: as they have far greater financial wealth than their European counterparts, even a small improvement in returns compounds to give the largest per capita wealth gain in our sample – around EUR24,800 in nominal terms and EUR15,000 in real terms.

12 Average per capita wealth, regardless of level of financial literacy; including currency and deposits, listed and unlisted shares, investment fund shares, and insurance and pension assets.

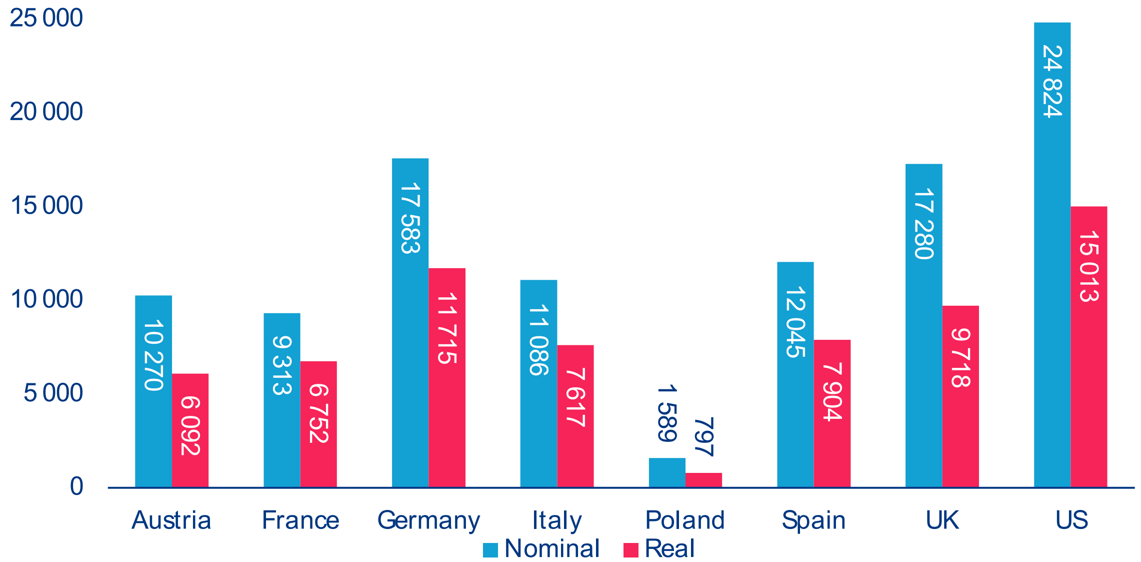
**Figure 15:** More deposits, more foregone return , average annual nominal returns (2005 – 2025) and share of bank deposits (2025) by country, in %




Source: Allianz Research

**A single reallocation, held for 20 years, adds up.** Nominal or real, the figures point in the same direction: the cost of excess deposits is large, and it grows with every passing year. Improving financial literacy does not only shape portfolio composition, it can have a meaningful and lasting impact on long-term wealth accumulation. Yet as the high share of „I don't know" responses reminds us, the first obstacle lies in households not knowing how to allocate their savings, leaving potential returns on the table.

**Figure 16:** The wealth gap, nominal and real gains from a portfolio shift, in Euro per capita




Source: Allianz Research

# Box 2: From awareness to action: the case for portfolio reallocation among risk-averse households

**Financial knowledge does not automatically translate into better financial decisions.** Our survey shows that even households with relatively strong financial knowledge continue to hold portfolios heavily tilted towards cash. Among respondents who prioritize preserving capital and securing stable retirement income (64.9%), the dominant investment response is not greater diversification but higher cash holdings. Bank deposits are the single most frequently chosen asset, selected by 38.7% as their primary intended allocation, suggesting that many households equate lower risk with holding more cash rather than constructing a more diversified and efficient low-risk portfolio. Mapping these investment intentions onto a simplified three-asset framework implies an allocation of approximately 41% cash, 38% equities and 21% bonds, compared with an average European household allocation of 34% cash, 45% equities and 21% bonds.

**The main barrier is behavioral, not informational.** Almost 70% of risk-averse respondents correctly answer the survey's inflation question, the highest share of any respondent group. Yet those who understand inflation are more, not less, likely to maintain large cash balances (19.3% vs 13.1%). This suggests that the persistence of cash-heavy portfolios reflects behavioral rather than informational barriers. Our findings point to three distinct behavioral personas that arrive at similar cash-heavy portfolios through different decision-making processes, each requiring a different approach to improve financial outcomes.

**Persona 1 – Liquidity seekers: Excess cash reflects precaution, not poor financial literacy.** More than half of cash-heavy households (56%) expect living costs to rise (78.5%), fear their income will fail to keep pace (69.0%) and anticipate drawing on savings to finance future spending (26.3%). Their preference for cash therefore reflects precaution rather than a misunderstanding of investment principles. Their portfolio challenge is not excessive risk-taking, but distinguishing between emergency savings and long-term investments. From a portfolio perspective, moving closer to the efficient frontier would achieve either higher expected returns at the same level of portfolio volatility or lower portfolio volatility for the same expected return by replacing part of excess cash holdings with high-quality bonds.

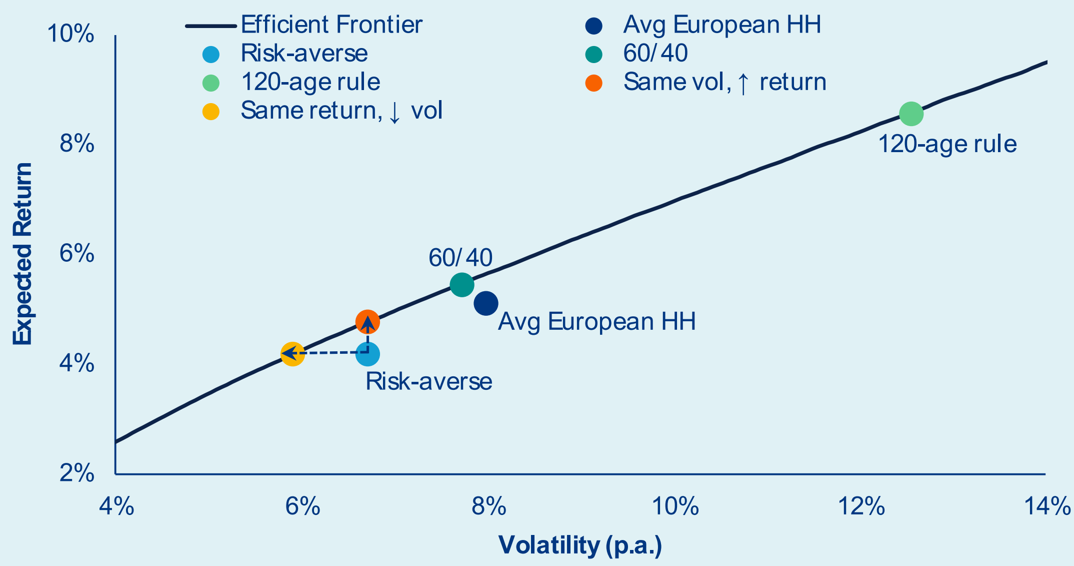
**Persona 2 – Procrastinators: Knowing there is a retirement gap does not guarantee action.** Nearly two-thirds (64%) expect a moderate or significant pension gap but continue to rely primarily on bank deposits. Unlike Persona 1, the issue is not immediate liquidity needs but delaying investment decisions, allowing today's portfolio to become tomorrow's default. Their portfolio challenge is not recognizing the retirement gap, but acting on it early enough to benefit from long investment horizons. Gradually reallocating long-term savings from excess cash into a more diversified mix of bonds and equities would better align their portfolios with those horizons.

**Persona 3 – Inertia investors: Many households stay in cash because inertia becomes the default.** Around one-fifth of cash-heavy households (19%) cannot be explained by either liquidity concerns or retirement expectations. Unlike Persona 2, they have not consciously decided to postpone investing – they simply remain with the default allocation, reflecting inertia rather than conviction. Their portfolio challenge is not selecting the optimal allocation, but moving away from an inefficient default. For many investors, a broadly diversified portfolio, such as a traditional 60/40 equity-bond allocation, would represent a more appropriate long-term starting point than maintaining excessive cash balances.

**Despite different behavioral motivations, all three personas converge on the same portfolio conclusion: replacing excess cash with bonds increases long-term wealth without increasing portfolio volatility.** The efficient frontier analysis quantifies the cost of excess cash holdings. Using estimated parameters from historical monthly returns, annualized at 2.5% Euro-area inflation and 100,000 simulated portfolios, both Pareto-efficient portfolios allocate zero cash, confirming that at the risk-averse group's volatility level (around 6.7% annualized), cash provides negligible volatility reduction while generating near-zero real returns[13]. Over the full period from 2002 to 2025, these more efficient portfolios compound to approximately 1.80–1.95× real wealth, compared with 1.65× for the observed risk-averse portfolio – a gain of 15–30 percentage points, driven primarily by eliminating the long-term drag from excessive cash holdings. Notably, the more efficient portfolios also hold less equity (27–33% vs 38%), illustrating that improving outcomes does not require taking more equity risk, but constructing a more efficient bond allocation (Figures 17 and 18).

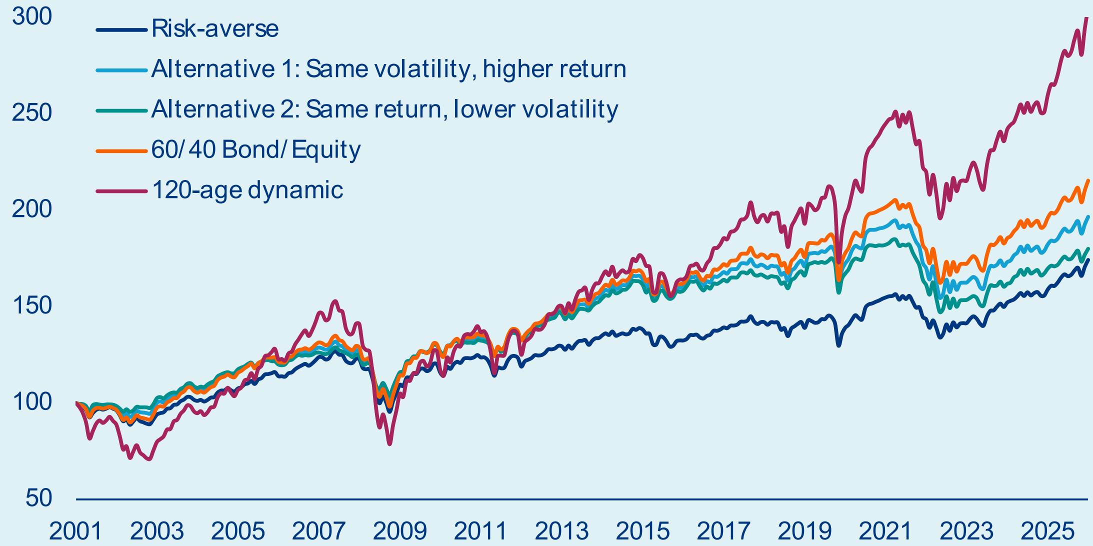
13 Pareto-efficient portfolios lie on the efficient frontier, meaning no other portfolio offers a higher expected return for the same level of risk or a lower level of risk for the same expected return.

**Figure 17:** Efficient Frontier: Improved portfolio allocation for a risk-averse household, expected return and volatility (p.a.)




Sources: LSEG Workspace, Allianz Research

**Figure 18:** Wealth accumulation over time: Cash-drag with significant underperformance, real return performance index (base year 2001 = 100)




Sources: LSEG Workspace, Allianz Research

**The same conclusion holds across the life cycle: reducing equity exposure with age is appropriate, but replacing it with cash rather than bonds is not.** Cash hoarding rises steadily with age, increasing from 8.8% among respondents aged 18–35 to 18.1% among those aged 66 and above, with an average hoarder being 53 years old. This pattern is consistent with standard life-cycle investing principles. The widely used 120-age rule recommends reducing equity exposure over time, implying around 71% equity for the average survey respondent aged 49, falling to well below 50% for individuals in their late 60s and 70s. Historical simulations confirm that a life-cycle portfolio following this rule delivers the strongest cumulative real wealth accumulation over the full sample period, but also the largest drawdowns – including losses of around 40% during the Global Financial Crisis – making it appropriate only for younger, risk-tolerant investors. For households approaching retirement, reducing equity exposure is entirely appropriate; the mistake is replacing equities with cash, rather than high-quality bonds, which provide a more efficient balance between capital preservation and long-term wealth accumulation (Figure 19).

**The portfolio solution is universal, but the behavioral barriers are not.** Across all three personas, replacing excess cash with high-quality bonds improves long-term retirement outcomes. What differs is not the portfolio recommendation, but how households can be helped to implement it. Precautionary savers need reassurance that investing surplus savings does not compromise their financial buffer. Passive planners need personalized projections and trusted advice that connect today's savings choices with tomorrow's retirement income. Default investors need better investment defaults rather than another decision to make. The broader implication is that improving household wealth requires more than financial education alone. It requires decision environments that help people translate financial knowledge into action.

**Figure 19:** Performance during crisis: Cash-hording does not protect from losses, real return performance index (base year = 100)

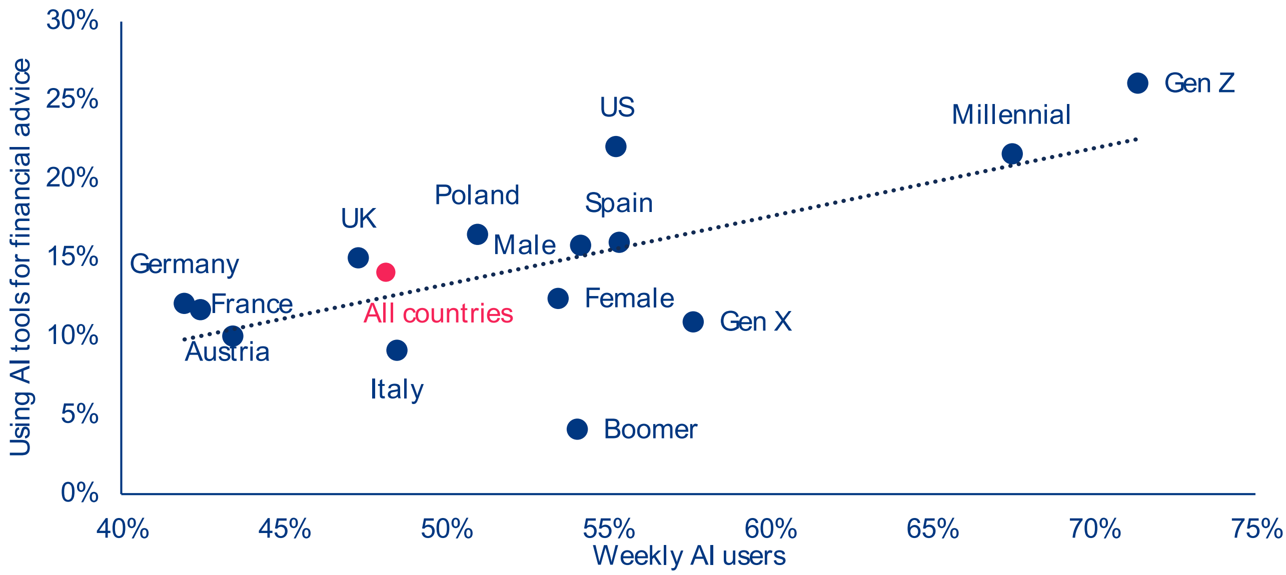

Sources: LSEG Workspace, Allianz Research

# From advisor to algorithm: the new financial advice ecosystem

**Financial advice is undergoing a quiet transformation.** A generation ago, households turned primarily to banks, professional advisors, the financial press or family when making important financial decisions. Those channels remain important today. But two new entrants have rapidly joined the ecosystem: financial influencers, who reach millions through social media, and artificial intelligence, which places an always-available assistant behind a simple prompt.

**AI is already mainstream, but its use for financial advice remains limited – for now.** Only 14.1% of respondents name AI as one of their two main sources of financial guidance, despite almost half (48.1%) using AI at least weekly and more than three-quarters of Gen Z (77.1%) doing so. This suggests that most people still draw a clear line between asking AI general questions and trusting it with their money. That line is blurring fast among the young: more than one in four Gen Z

**Figure 20:** Weekly AI use vs. using AI for financial advice, by country, gender and generation, share of respondents in %.




Source: Allianz Research

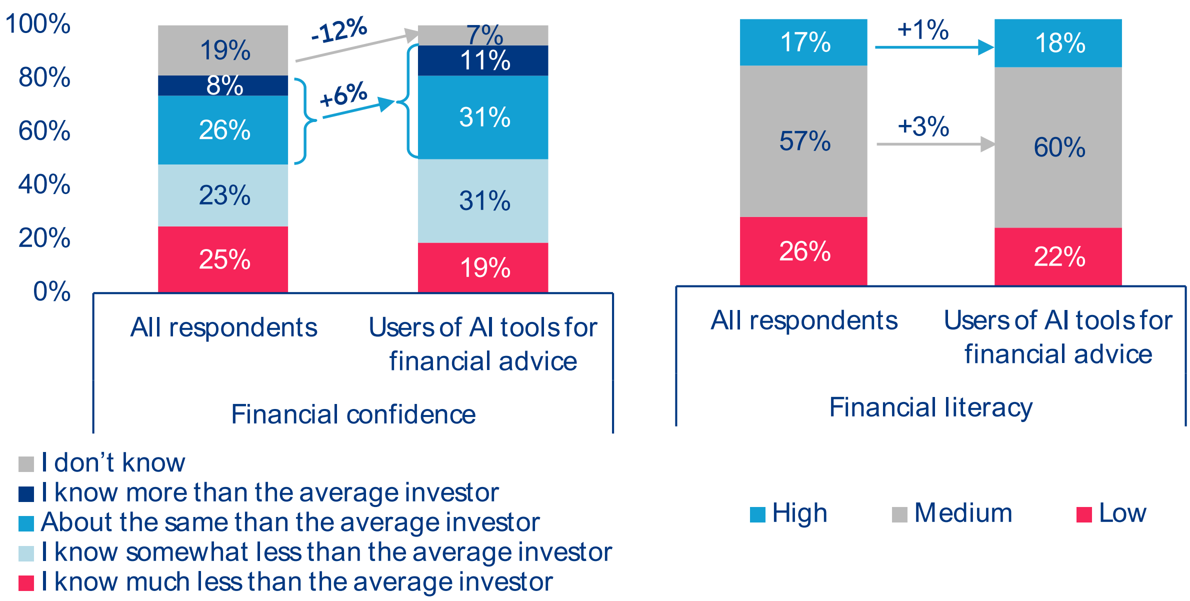
respondents (26.1%) and more than one in five Millennials (21.6%) already count AI among their main advice sources, rising to 30.9% of young men – and past 40% in Poland and nearly 38% in the US – compared with only 22.7% of young women. Financial influencers reach almost exactly the same demographic, so that for a growing share of young households the advice ecosystem is now as much TikTok, podcasts and chatbots as a bank or a personal advisor.

**The most striking finding is a mismatch between two things that ought to move together, but don't:** AI users are far more confident about their financial knowledge, yet not significantly more financially literate (Figure 21). On confidence, the gap is wide and consistent. 42.5% of AI users rate their financial knowledge as average or above average, compared with 33.3% of all respondents. Only 18.8% of those who use AI for financial advice feel they know "much less" than the average investor against 25.0% of all respondents, while the share believing they know more rises from 7.6% to 11.4%. Uncertainty falls sharply too: just 7.4% of AI users answer "I don't know," compared with 18.7% of the full sample.

**Measured literacy, however, differs little: 18.1% of AI users reach the high-literacy band, compared with 17.5% of the overall sample (Figure 21).** The confidence premium therefore far exceeds the literacy premium. Whether AI genuinely deepens understanding or merely inflates self-assurance cannot be settled from the survey, but the mismatch deserves attention: financial decisions are shaped not only by what people know, but by what they believe they know. AI can make financial information easier and cheaper to access, but it cannot substitute for the financial literacy needed to assess that information and make better financial decisions. This creates the risk of overconfidence if users become more certain without becoming better equipped to judge the quality of the advice they receive.

**This does not mean AI is displacing professional advice so much as splitting along the literacy divide.** For financially literate households, AI can complement professional advice, making information more accessible and easier to process. For less literate households, however, it may instead serve as a substitute for professional advice seen as expensive, inaccessible or intimidating[14]. That is where the risk concentrates: the users leaning hardest on AI, trusting it most, and least able to judge its answers are disproportionately those with the least financial knowledge. The distinction will be one of the defining questions for financial inclusion over the coming decade. AI can democratize access to financial information at unprecedented scale, but access alone does not create financial capability. The task for policymakers, educators and AI providers alike is to ensure AI does not merely make advice more available, but helps people make better decisions.

**Figure 21:** Confidence and financial literacy, AI users vs. the overall sample, share of respondents in %.




Note: Confidence indicators move sharply in AI users' favor while literacy indicators do not – the confidence premium exceeds the literacy premium. Source: Allianz Research.

14 Recent research on AI financial advice supports this distinction: AI-generated financial advice can move household saving, spending and investment decisions closer to standard economic benchmarks, but outcomes depend on users' financial literacy and ability to formulate effective prompts. More financially literate users received advice associated with higher simulated wealth, while less financially literate users fare worse. The authors conclude that regulators and fintech platforms might need to address disparities in AI-driven advice. See: Choukhmane, T., T. de Silva, W. Lin, and M. Akuzawa (2026): "AI Financial Advice: Supply, Demand, and Life Cycle Implications", https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6446286

# What can be done: a five-point financial literacy playbook

**Our follow-up survey, three years on, delivers an uncomfortable verdict: financial literacy has improved little, while the need for it has never been greater.** While the responsibility for financial security does not sit with households alone, people are being asked to take greater responsibility for their financial future, and need the tools and knowledge to make informed decisions. Financial literacy is therefore no longer simply about understanding financial concepts – it has become a prerequisite for building, protecting and growing wealth over a lifetime. Yet awareness campaigns and one-off initiatives have not moved the needle. Closing persistent literacy gaps requires a new policy playbook.

**Point 1: Build financial capability throughout life.** Financial capability should be treated as a lifelong skill, embedded in financial education across school curricula while expanding opportunities for adult learning to keep up with evolving financial products, pension systems and digital technologies. Employers should reinforce these foundations through workplace financial education, while financial institutions can support learning with accessible educational content and practical decision-making tools. One example is the Allianz School for Life, a free, open-access digital learning platform designed to make financial and risk education widely accessible[15]. The objective is not simply higher test scores, but equipping people with the knowledge required to navigate increasingly complex financial decisions throughout their lives.

**Point 2: Make future outcomes visible.** Knowing financial concepts is rarely enough if people cannot see the consequences of today's choices. Tools to support decision-making should therefore become part of the financial infrastructure. Governments can accelerate the rollout of pension dashboards and personalized retirement projections. Employers can complement these with workplace pension guidance and retirement planning. Financial institutions should provide digital tools that illustrate how different saving and investment decisions affect future retirement income and long-term wealth. Making future outcomes tangible is likely to change behavior more effectively than providing additional information alone.

**Point 3: Turn knowledge into action.** Our findings show that the greatest obstacle is often not ignorance but inertia. Many households understand basic concepts, yet continue to hold inefficient, cash-heavy portfolios, or postpone investment decisions altogether. Policies therefore need to make good decisions easier. Employers can use automatic enrollment, regular pension reviews and financial wellbeing programs to encourage action. Financial institutions should simplify products, improve defaults, reduce unnecessary complexity and provide behavioral nudges that help households gradually move from saving, to long-term investing.

**Point 4: Build confidence – and healthy skepticism – in the AI age.** Financial literacy increasingly means knowing how to evaluate advice, not simply where to find it. AI has dramatically lowered the cost of accessing financial information, but not necessarily the ability to judge its quality. Financial education therefore needs to help people calibrate their confidence, recognize when professional advice is required and understand AI's strengths and limitations. Policymakers, educators, employers, financial institutions and AI providers all have a role in promoting AI as a complement to trusted professional advice, rather than using it as a substitute.

**Point 5: Reach everyone – but focus where the gaps are largest.** Financial literacy policies should be universal in reach, but targeted in delivery. The largest deficits identified in this report are concentrated among women, younger generations and adults with lower educational attainment. These groups require interventions tailored to their circumstances: early financial education before major financial decisions arise; workplace support at key life events; advice and investment tools designed to increase confidence among women; and accessible, practical learning opportunities for adults with lower educational attainment. Universal access ensures broad financial capability while targeted delivery helps close persistent inequalities.

15 See: Allianz School For Life and Allianz MoveNow

# Appendix A : Survey summary statistics

The survey queried more than 8,000 respondents across eight countries: Austria, France, Germany, Italy, Poland, Spain, the United Kingdom and the United States. It was conducted as a web-based survey between 22 April and 25 May 2026. Overall responsibility lies with Allianz Research, Allianz Investment Management SE. Sample planning and fieldwork were carried out by Qualtrics.

Respondents were selected using representative quota sampling. Qualtrics was given quotas specifying how many people to survey in each country and which selection criteria to apply, with quotas set in line with official statistics for gender and age groups. A comparison with official statistics confirms that the resulting sample broadly corresponds to the population aged 18 and over in each of the eight countries.

Appendix Table 1: Sample statistics

| | Total | Austria | France | Germany | Italy | Poland | Spain | UK | USA |
|---|---|---|---|---|---|---|---|---|---|
| Number of respondents | 8048 | 1006 | 1006 | 1006 | 1006 | 1006 | 1006 | 1006 | 1006 |
| Gender, in % of respondents | | | | | | | | | |
| Female | 50.6% | 50.0% | 51.4% | 49.9% | 50.7% | 50.9% | 50.4% | 50.9% | 51.0% |
| Male | 49.2% | 50.0% | 48.2% | 49.9% | 49.1% | 49.0% | 49.4% | 48.9% | 48.9% |
| Age groups, in % of respondents | | | | | | | | | |
| Gen Z | 15.5% | 15.4% | 16.1% | 13.6% | 13.1% | 15.2% | 14.3% | 16.9% | 19.6% |
| Millennial | 25.9% | 26.8% | 24.8% | 25.8% | 21.9% | 27.0% | 24.5% | 28.3% | 27.8% |
| Gen X | 31.5% | 34.1% | 30.2% | 26.2% | 35.8% | 33.6% | 41.1% | 25.0% | 25.6% |
| Boomer | 27.2% | 23.7% | 28.9% | 34.3% | 29.3% | 24.3% | 20.2% | 29.8% | 27.0% |
| Education, in % of respondents | | | | | | | | | |
| Primary (less than high school) | 4.8% | 2.2% | 3.7% | 2.6% | 14.1% | 4.0% | 2.8% | 2.0% | 7.2% |
| Secondary (high school) | 52.5% | 75.2% | 47.9% | 69.2% | 52.5% | 55.2% | 36.0% | 43.6% | 40.1% |
| Tertiary (university or comparable) | 42.7% | 22.6% | 48.4% | 28.2% | 33.4% | 40.9% | 61.2% | 54.4% | 52.8% |

| Generation definition | Birth years |
|---|---|
| Gen Alpha* | 2013 - 2031 |
| Gen Z | 1997 - 2012 |
| Millennial | 1981 - 1996 |
| Gen X | 1965 - 1980 |
| Boomer | 1946 - 1964 |
| Silent** | 1928 - 1945 |

* Not part of the survey sample.
** Not mentioned in the analysis as it was a very small portion of respondents.

# Appendix B: Financial literacy questions

**1.Understanding of simple interest rate.** Suppose you had USD500 in a savings account and the interest rate was 5% per year. After one year, how much do you think you would have in the account if you left the money to grow?

**a.USD 525**
b.USD 555
c.USD 532
d.I do not know

**2.Understanding of compound interest rate (numeracy).** Suppose you had USD1000 in a savings account and the interest rate was 2% per year, compounded annually. After 5 years, how much do you think you would have in the account if you left the money to grow?

**a.More than USD 1100**
b.Exactly USD 1100
c.Less than USD 1100
d.I do not know

**3.Understanding of inflation.** Imagine that the interest rate on your savings account was 5% per year and inflation was 7% per year. After 1 year, how much would you be able to buy with the money in this account?

a.More than today
b.Exactly the same
**c.Less than today**
d.I do not know

**4.Understanding of risk diversification.** Please tell me whether this statement is true or false. "Buying a single company's stock usually provides a safer return than a mutual fund."

a.True
**b.False**
c.I do not know

**5.Lotteries.** Lottery A pays a prize of USD200 and the chance of winning is 5%. Lottery B pays a prize of USD90,000 and the chance of winning is 0.01 %. In either case, if one does not win, one does not get any money. Which lottery pays the higher average amount? (Please pick one option only)

**a.Lottery A**
b.Lottery B
c.These two lotteries pay the same average amount
d.I do not know

**6.You can invest in two projects.** Project A will either deliver a return of 10 or 6%, with either outcome equally likely. Project B will either deliver a return of 12 or 4%, with either outcome equally likely. Which of the following is true? Compared to Project B, Project A has… (Please pick one option only)

a.Higher return and lower risk
**b.Same average return and lower risk**
c.Lower return and higher risk
d.I do not know

**7. You want to set aside money today for certain expenses in two years.** Which of the following investment offers are suitable:

a.A fund with an annual return of 7% in the past but also large fluctuations
**b.A two-years fixed-term deposit with a fixed interest of 1.5% per annum**
c.I do not know

8.How would you calculate financial net worth?

**a.Assets minus liabilities**
b.Liabilities minus assets
c.Assets plus liabilities
d.Assets divided by liabilities
e.I don't know

**9.Imagine we are throwing a loaded die (6 sides).** The probability that the die shows a 6 is twice as high as the probability of each of the other numbers. On average, out of 70 throws how many times would the die show the number 6?

**a.20 out 70 throws**
b.30 out of 70 throws
c.35 out of 70 throws
d.I don't know

# Allianz School for Life


**Allianz School For Life is a free, open-access digital learning platform designed to make financial and risk education widely accessible because better decisions start with better knowledge.** The platform offers learning journeys covering budgeting, investing and risk management, tailored to different life stages – from children and teenagers to young adults – and designed to make financial education engaging, accessible and relevant to everyday life. Beyond digital learning, Allianz also supports on-the-ground delivery through local programs, partnerships and community initiatives, helping ensure that financial education reaches people wherever they are.

**Check it out:**

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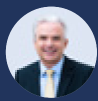

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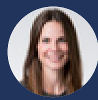

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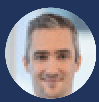

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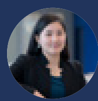

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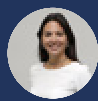

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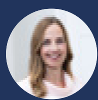

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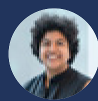

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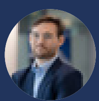

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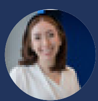

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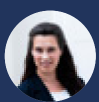

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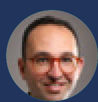

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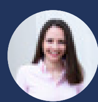

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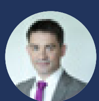

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Allianz Research | 29 July 2026

# Cracking finance: Why the quantum threat could arrive before the reward


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## In Summary


- **Most technologies create value before they create problems, but quantum computing may break that pattern.** Imagine two clocks ticking side by side: one measures the time until quantum computers become economically useful, the other the time until they can break the cryptography that underpins the financial system. Which clock strikes first is a genuinely open race.
- **The upside potential is real but narrowly concentrated around derivative pricing and other complex financial simulations. Beyond these areas, its impact on the financial industry is likely limited.** Quantum computing accelerates Monte Carlo methods behind financial simulations, while it shows no consistent advantage yet over improving classic methods for portfolio optimization and financial machine learning. Insurers are well positioned as simulation and optimization are at the heart of liability valuation, risk pricing, capital modelling and asset-liability management. Quantum money could use uncopiable quantum states to authenticate value and prevent counterfeiting, although commercial deployment remains remote. The rewards remain theoretical for now, threatened by error-correction overhead and a rising classical benchmark, until large machines exist. But quantum-inspired methods deliver operational value on classical hardware today, and the talent, tooling and know-how built along the way will carry over.
- **Q-day could come sooner than we think: data harvested today can be decrypted later.** The recipe for breaking today's encryption has existed since 1994 and waits only for the machine, which is also the smaller one, needing roughly 1,250–1,450 logical qubits vs around 4,700–7,500 for leading financial applications. Health records, classified government information and sensitive financial data must stay confidential for decades, but experts put the probability of Q-day, the point at which a cryptographically relevant quantum computer exists, arriving within ten years at 28–49%. Defensive standards have existed since 2024, and regulators have set dates: The US National Institute of Standards and Technology (NIST) will phase out today's algorithms after 2030, the EU requires critical financial infrastructure to migrate by the same year.
- **Digital assets make the vulnerability visible: USD400bn of Bitcoin is already at risk, and that is just the tip of the iceberg.** By spring 2026, roughly 6m bitcoins, 30% of supply, sat behind exposed public keys, and Ethereum extends the exposure to stablecoins, bridges and tokenized assets. Spillover to mainstream finance would be systemic: The BIS puts losses above 1% of GDP over 15–20 years, while a severe Hudson Institute scenario reaches up to USD3.3trn in GDP losses (3% of global GDP) over 6 quarters.
- **Both private and public sectors are visibly chasing the reward, while concerns about the threat play out mostly behind closed doors.** Investors bet mostly on the hardware: about 97% of each dollar funds the machine and under 1% the applications, on an estimated USD12.6bn raised by quantum startups in 2025. Public efforts reveal a geographic imbalance: The US prepares for reward and threat via parallel executive presidential orders, while Europe, which draws just 5% of private quantum investment versus over half for the US, pursues its own response through fragmented policy instruments, risking a repeat of its AI experience as adopter and regulator of a technology it does not own. Preparation for the threat is purely an internal IT budgeting decision, competing resources with AI, and only about 35% of large organizations completed a cryptographic inventory. Institutions need crypto-agile architectures, prioritized inventories, migration roadmaps and tested fallbacks before the clock reaches zero: preparing for only one clock is the wrong strategy, regardless of which strikes first.

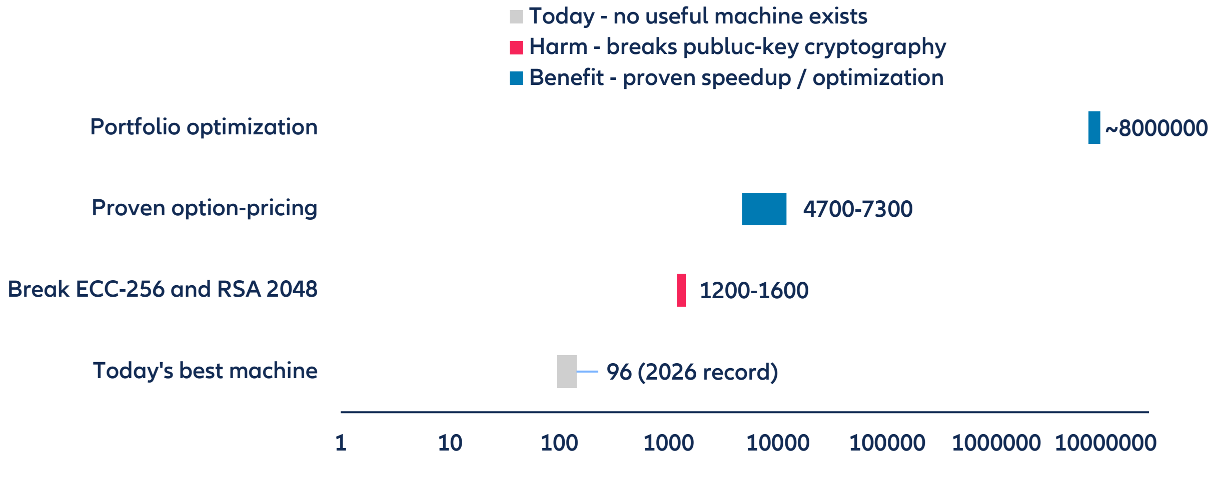
## Two clocks ticking at different speeds

**Most technologies create value before they create problems.** The automobile delivered mobility before congestion became a policy concern; the internet created a digital economy before it created cybercrime. Even when the benefits arrived slowly – electrification took decades to lift factory productivity, and computers were once, as Robert Solow famously observed, visible "everywhere but in the productivity statistics" – the direction was the same: the rewards came first, and the risks followed. Quantum computing may break that pattern. Imagine two clocks ticking side by side. One measures the time until quantum computers become economically useful; the other, the time until they can break the cryptography that underpins the financial system. There is a credible case that the threat clock is running at least as fast as the reward clock, and that institutions, drawn to the visible upside, are quietly preparing for only one of the two.

**To see why the two clocks can run at different speeds, start with the machine itself.** A quantum computer is not simply a faster version of a conventional computer, nor does it "try every answer at once". It is a different class of machine that can solve a narrow set of mathematical problems in ways that remain intractable for conventional systems. Its basic unit is the qubit, the quantum counterpart of the classical bit. Whereas a conventional bit must be either 0 or 1, a qubit can occupy a quantum state that allows information to be processed differently. The difficulty is that qubits are fragile. The physical qubits built in today's laboratories are sufficiently error-prone that long computations quickly dissolve into noise. Useful large-scale quantum computing therefore depends on error correction: bundling large numbers of physical qubits into a much smaller number of stable logical qubits. Logical qubits, rather than raw physical-qubit counts, are the more meaningful measure of what a machine can actually compute. Even then, quantum advantage is highly specific. Most operations performed by banks, insurers and asset managers would see little benefit from quantum hardware. Yet one of the problems for which quantum computers are exceptionally well suited is factoring and discrete logarithms, the mathematical foundations of much of today's public-key cryptography.

**On paper, and only against the most space minimized estimates, the code-breaking machine can be the smaller one.** Attacking the elliptic-curve cryptography behind a Bitcoin key has been estimated to require roughly 1250–1,450 logical qubits, depending on circuit design and the trade-off between qubit count and runtime. The lowest estimates for breaking RSA-2048, a distinct problem from the elliptic-curve case, have fallen to a similar order of magnitude as architectures and algorithms have improved, though the standard single-shot figure stays several times higher. By comparison, a leading fault-tolerant proposal for pricing and risk using amplitude estimation requires roughly 4,700–7,500 logical qubits, while a general and economically relevant portfolio-optimization advantage may require far more resources and has not yet been demonstrated (Figure 1).

Figure 1: Estimated number of logical qubits required for different tasks




*Sources: Babbush et al. (2026), Gidney (2025), Gong et al. (2026), Dalzell et al. (2023), Allianz Research. Note: The estimation of the number of logical qubits for portfolio optimization comes from a study using a method called the quantum interior point method (QIPM). Under this method, the number of qubits scales as $800n^2$, where n is the number of assets in the portfolio.*

**A simple count of logical qubits, however, does not settle the matter. Gate counts, fidelities and circuit depth matter as much as qubits and the deeper asymmetry is competitive rather than technical.** For breaking cryptography, a quantum computer is essentially the only tool that can do the job; there is no classical shortcut, so the code-breaking machine only must exist. For finance, quantum must compete: classical hardware and algorithms already perform these calculations and keep getting better, so even a machine that is smaller on paper must clear a far higher bar to be worthwhile. Cryptographic attacks thus appear to be among the smallest clearly specified fault-tolerant applications, while the best-developed financial use cases demand larger and faster machines and must still beat a moving classical benchmark that the attacker never faces. The same asymmetry shapes the economics of building the machines: money spent chasing a financial edge competes against cheaper classical improvements, whereas money spent on defense buys protection against something only quantum can do.

**The decisive asymmetry, though, is about recipes rather than machines.** Shor's algorithm, the quantum procedure published by Peter Shor in 1994 that efficiently factors large numbers and solves the related discrete-logarithm problem on which public-key cryptography rests, has existed already for three decades. It is proven, published and has a clearly defined target: given sufficiently capable fault-tolerant hardware, it can recover the private keys behind the public-key cryptography, which authenticates and encrypts nearly every digital exchange, whether a bank transfer, a private message or a social network post. The remaining uncertainty is about the time when a machine with scale, speed and error correction sufficient to run the algorithm will exist.

**Financial quantum algorithms are not absent, but their route to economic value is far less settled.** Quantum amplitude estimation offers a firm mathematical basis for accelerating Monte Carlo calculations, while other approaches target optimization, machine learning and linear algebra. Yet for amplitude estimation the potential gain is quadratic rather than exponential, and for optimization and machine learning many claimed advantages remain heuristic or highly problem specific**.** The destructive use case already has its recipe and waits mainly for the machine. Productive uses need two things at once: the very hardware the adversary is also waiting for, and financial algorithms that still have to be found, implemented and shown to outperform strong classical methods. However, the two clocks compete on different dimensions, both worth exploring, and it is hard to say which clock will strike first.

## The reward clock

**The strongest case for quantum computing in finance starts from a simple observation: the industry already devotes enormous computing capacity to simulating uncertainty.** Derivative pricing, market-risk measurement, economic-capital calculations and stress testing rely heavily on Monte Carlo methods – repeated simulations of thousands or millions of possible future outcomes. Quantum amplitude estimation can, in principle, achieve a given level of statistical accuracy with fewer evaluations than classical Monte Carlo. The potential result is not a wholly new calculation but the ability to perform existing calculations faster, more frequently or at greater resolution: compressing overnight risk runs, refreshing capital positions more often and evaluating a broader range of scenarios. Because Monte Carlo methods are used for a variety of applications, like pricing complex derivatives, calculating Value-at-Risk and expected shortfall, estimating counterparty exposure or measuring economic capital, the industry is unusually well aligned with one of the few quantum applications supported by a firm mathematical speedup.

**The gain should nevertheless not be overstated.** A quadratic acceleration can be valuable, especially for deeply nested simulations, but the moving classical benchmark takes concrete form here: specialized chips, GPUs, distributed cloud infrastructure, variance reduction and continually improving classical algorithms. A commercially useful quantum system must outperform them after incorporating error correction, data loading, state preparation and the cost of the full workflow. This is the single most important test any financial quantum application has to pass, and it is why even a genuine speedup on paper may not translate into a genuine advantage in practice.

**Insurers sit at a competitive intersection: the one quantum advantage with the strongest mathematical foundation maps directly onto both their heaviest regulatory calculations and parts of their investment process.** On the underwriting and capital side, valuing life, health and long-tail property-and-casualty liabilities can require simulations extending decades into the future. Those valuations feed reserving, stress testing, economic-capital calculations and Solvency II coverage ratios, often through layers of nested simulations that become computationally expensive. Amplitude estimation targets exactly this kind of workload: best-estimate liability valuation, solvency-capital and Value-at-Risk computation and the economic-scenario generation that feeds them. Faster simulation could improve efficiency, but more importantly it could change the cadence of risk management, moving capital and solvency monitoring from a periodic exercise toward more frequent updates and more dynamic steering and hedging. The same techniques carry over to the asset and asset-liability management side. Evaluating a proposed allocation across thousands of economic scenarios, generating those scenarios, and measuring tail and counterparty risk all belong to the same simulation-intensive family that amplitude estimation accelerates.

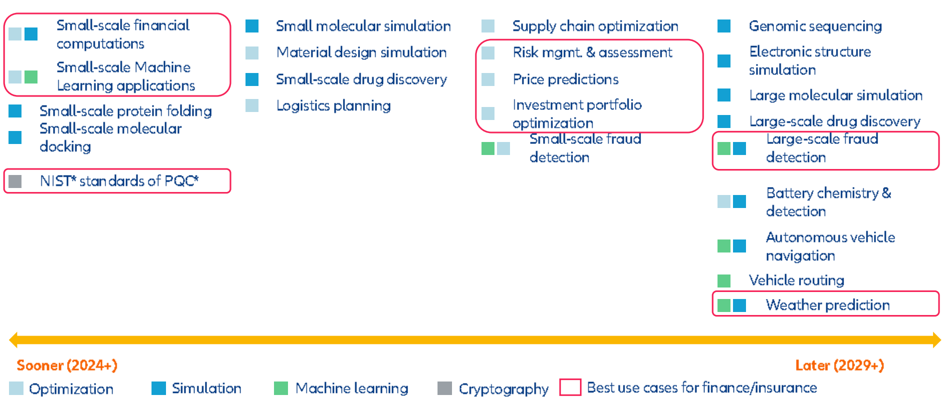
**This is also where the wider industry is concentrating its experiments.** Goldman Sachs has explored quantum Monte Carlo for derivative pricing; JPMorgan has tested optimization routines for portfolio and collateral problems and Crédit Agricole has piloted hybrid credit-risk models. The honest split is that evaluating an allocation is the part with a proven speedup, whereas choosing the optimal allocation is a combinatorial-optimization problem for which quantum methods have yet to show a robust advantage. The credible opportunity therefore lies first in evaluating allocations faster and with richer scenarios, rather than in expecting a quantum machine to discover the optimal portfolio automatically.

**Beyond simulation, the list of proposed applications is long, but the evidence base less so**. Corporate and investment banks are testing collateral allocation, balance-sheet optimization and faster pricing systems. Asset managers are exploring portfolio construction, asset modeling and scenario analysis. Payments businesses see potential applications in fraud detection, compliance and, further out, quantum-secure settlement. Retail banks may eventually use quantum or quantum-inspired methods for credit assessment and large-scale decision problems (Figure 2).

**At the most experimental end lies quantum money.** First proposed by Stephen Wiesner and later developed by Charles Bennett and Gilles Brassard, the concept does not describe a new currency comparable with cryptocurrencies or central-bank digital money, but rather payment tokens encoded in quantum states that cannot be perfectly copied without disturbing them. In principle, quantum mechanics could therefore provide a new way of authenticating value and preventing counterfeiting. Literal quantum banknotes remain technologically remote, however, as they would require fragile quantum states to be stored, transferred and verified reliably, while secure public verification remains an open research problem. More recent experiments have shifted toward quantum payment tokens and one-time cryptograms, using quantum communication to make payment authorization harder to forge without requiring users to preserve a quantum state indefinitely. These approaches remain far from commercial deployment, but they illustrate a broader potential reward.

**Most of these opportunities are less certain than the Monte Carlo case**. Optimization and machine-learning applications face intense competition from classical solvers that have improved rapidly over the past decade, and a result demonstrated on a small or synthetic problem is not necessarily relevant to an institution operating with noisy data, latency constraints, regulation and legacy infrastructure. The likely economic impact is therefore concentrated in a narrower set of applications than the long lists of potential use cases suggest.

Figure 2: Quantum use cases by expected time to maturity




*Sources: Allianz SE Cybersecurity & NextGen IT Think Tank, Arthur D. Little, Allianz Research*

**Even within that narrower set, value is available today with no quantum hardware advantage**. The most dependable gain comes from reformulating the model itself. When you take an actuarial, Asset-Liability-Management or capital model and rewrite it as a QUBO, Ising or tensor form[1], you force an explicit restatement of variables and constraints, surfacing stale constraints and dead risk factors whose corrections feed back into classical production models. Named quantum-inspired solvers are already in commercial use: Multiverse Computing and Crédit Agricole CIB report tensor-network pricing that matches accuracy with fewer parameters and faster training, and Toshiba's Simulated Bifurcation Machine has run low-latency pairs-trading at the Tokyo Stock Exchange. Independent benchmarks find plain simulated annealing comparable in some regimes, and the edge of these solvers comes from specialized hardware rather than a quantum effect. None of this proves a speedup against the moving classical benchmark; the value is real but operational.

**In fact, the reward clock has two horizons.** A nearer horizon already exists in hybrid quantum-classical workflows and "quantum-inspired" methods that borrow mathematical techniques from quantum computing while running on conventional hardware; platforms such as NVIDIA CUDA-Q couple GPUs with quantum processors and speed up circuit design, simulation and hybrid execution. These gains are operational rather than proof of a quantum speedup, but they lower experimentation costs and help institutions build talent, tooling and credible benchmarks. The far horizon is fault-tolerant quantum advantage: a machine that performs a complete financial task more efficiently than the best classical alternative. Putting a credible number on that long-run reward is not yet possible. The upside is that the reward can therefore arrive the way most technology rewards do: gradually, as capability, tooling and know-how spill over from experimentation into production, with modest gains already available today and a larger advantage possible later. The threat keeps no such schedule. A cryptographic break would not diffuse gradually through the economy, but rather arrive all at once, and its countdown has already begun. It is to that clock that we now turn.

## The threat clock

**What Shor's algorithm can break extends far beyond any single market.** RSA and elliptic-curve schemes secure key exchange, digital signatures, authentication and ownership across the financial system and, in fact, across the wider digital economy that finance depends on. Symmetric cryptography such as AES is also weakened by quantum search but can generally be protected by increasing key lengths; public-key infrastructure requires a more fundamental migration.

**The machine does not need to exist for the exposure to exist: adversaries can harvest now and decrypt later.** An adversary can collect encrypted information today and store it for future decryption once suitable hardware arrives, a practice known as "harvest now, decrypt later". For data that must remain confidential for a decade or more, like merger discussions, customer records, trading strategies and health information, the clock has effectively already started. Quantum threat is unusual because part of the future exposure sits in the past: information transmitted years before a cryptographically relevant machine exists may later become readable.

**Is Q-Day coming?** Expectations of when that machine arrives have moved materially. No general-purpose machine has yet run Shor's algorithm at a cryptographically meaningful scale, but recent studies have sharply reduced estimates of the physical and logical resources required to attack commonly used schemes, and expert opinion has shifted accordingly: The Global Risk Institute's 2025 *Quantum Threat Timeline Report*, answered by 26 experts, assigns a 28–49% probability to a cryptographically relevant quantum computer within ten years, depending on the definition used. The range is wide and should be treated as expert judgment rather than an objective probability, but it is sufficient to make early migration to quantum-safe infrastructure economically rational. Cryptographers formalize the logic as Mosca's inequality: whenever the time needed to migrate plus the required secrecy lifetime of the data exceeds the time until a capable machine arrives, the rational moment to start is already now. The cost of starting too early is a phased technology program; the cost of starting too late could be the loss of trust in critical infrastructure.

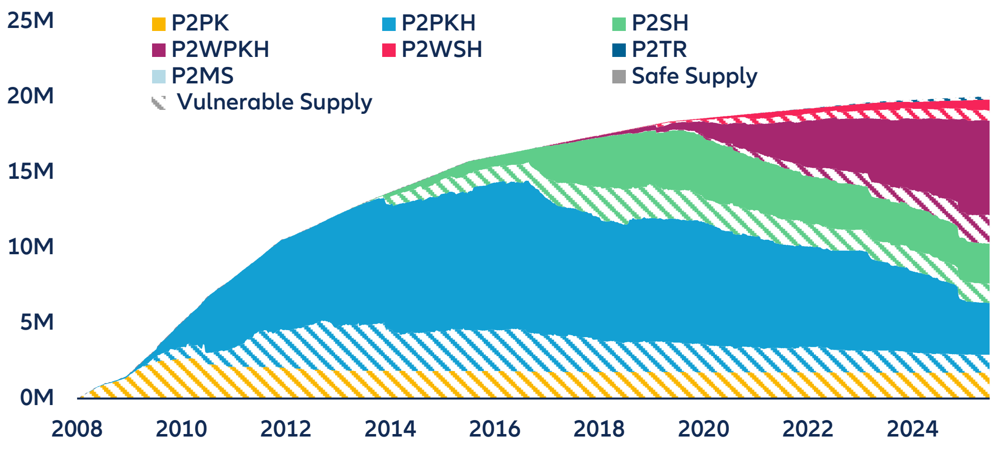
[1] Mathematical forms originally designed for quantum or quantum-inspired solvers, where QUBO stands for Quadratic Unconstrained Binary Optimization, essentially a way of expressing a problem as binary yes/no choices with trade-offs between them.

**The defensive mathematics is already settled: In August 2024, the US National Institute of Standards and Technology (NIST) finalized the first post-quantum cryptographic standards, including ML-KEM for key establishment and ML-DSA for digital signatures.** Regulators are attaching dates to the transition: NIST plans to deprecate today's vulnerable public-key algorithms after 2030 and disallow them after 2035, turning the migration deadline into a compliance fact rather than a physics forecast. The key uncertainty is therefore no longer whether replacement algorithms will exist, but whether institutions can identify, test and replace their cryptographic dependencies before the threat becomes operational. The new algorithms carry real engineering costs. Post-quantum public keys, ciphertexts and signatures are often larger than their current counterparts, creating challenges for bandwidth, storage, processing and constrained devices. When the Eurosystem and the BIS tested post-quantum cryptography in Project Leap, message sizes expanded and verification times increased. The increase was manageable in many applications, but material in high-volume or latency-sensitive infrastructure. Industry migration has begun in parallel: JPMorgan has operated a production quantum-secured, crypto-agile network since 2024.

**The lesson is organizational rather than scientific.** Institutions must inventory every system that relies on cryptography, identify data requiring long-term confidentiality, rank systems by risk and migration difficulty, test vendor readiness and build crypto-agility so that algorithms can be replaced without redesigning the entire application. This resembles a multi-year rebuild of trust infrastructure, not a conventional software patch.

**Public blockchains could show what a cryptographic migration looks like when no one is in charge of the migration.** Bitcoin provides the clearest illustration. By spring 2026, roughly 6mn Bitcoin, or around 30% of total supply, worth roughly USD400bn at current prices, sat behind public keys already visible on-chain. Some of that exposure is operational and can be mitigated by moving funds to fresh addresses. Some is structural. Early pay-to-public-key outputs and a large share of Satoshi-era coins reveal public keys by design, while the corresponding private keys are widely believed to be lost. The chart below (Figure 3) illustrates how that vulnerable supply has evolved across script types over time. The precise number is less important than the principle: a meaningful share of the world's largest digital asset already sits behind cryptography that a future quantum computer could challenge.

Figure 3: Bitcoin supply by script type and vulnerability




*Sources. Google Quantum AI, Allianz Research. Note: Labels refer to Bitcoin address types, from the oldest (P2PK, 2009–11) to the newest (P2TR "Taproot", since 2021). What matters is the split: safe supply (solid areas) are coins whose cryptographic key is still hidden and thus cannot be attacked; vulnerable supply (striped areas) are coins whose key is already visible on the blockchain and could in principle be stolen by a powerful future quantum computer. Glossary of the legend: P2PK = Pay-to-Public-Key (key always visible); P2PKH = Pay-to-Public-Key-Hash; P2SH = Pay-to-Script-Hash; P2WPKH / P2WSH = SegWit formats; P2TR = Pay-to-Taproot (key visible by design); P2MS = bare multisig (keys visible)*

**Ethereum broadens the issue from digital assets to digital infrastructure.** The exposure extends beyond user balances to administrative keys controlling stablecoins, tokenized assets, bridges and layer-two networks. As tokenization expands, the same cryptography that secures blockchains will increasingly secure representations of real world financial assets. What begins as a cryptocurrency issue therefore becomes a custody, settlement and investor protection issue for mainstream finance. Industry projections envision tokenized real world assets reaching trillions of dollars over the coming decade; those assets would inherit the same cryptographic vulnerabilities as the networks on which they reside.

## Systemic implications

**A quantum-enabled cryptographic systemic break would differ from most cyber events for which financial institutions currently plan.** Most attacks affect a particular firm, service or network and can eventually be contained and restored. A break in a widely used cryptographic method could spread much further because banks, payment systems and market infrastructures rely on the same small set of standards.

**The risk could extend well above the disclosure of confidential information.** If public keys and digital signatures could no longer be trusted, an institution might be unable to establish whether a certificate is genuine, whether a message came from the stated counterparty or whether a payment instruction was authorized. An attacker could potentially impersonate a customer or institution, alter instructions or distribute malicious software under an apparently valid signature. The problem would then move quickly from data security to the integrity of payments, settlement and ownership records.

**The systemic channel is confidence.** Central banks and financial institutions are connected through common payment, messaging and settlement infrastructures, all of which depend on reliable authentication. A problem at one institution would not automatically bring down the system, but uncertainty over whether messages and signatures remained genuine could cause counterparties to delay transactions or suspend activity while the breach was investigated. The stability of the financial system is fundamentally dependent on cryptographic security. Maintaining trust and confidence is therefore central to the response, not merely recovering the data or systems initially affected.

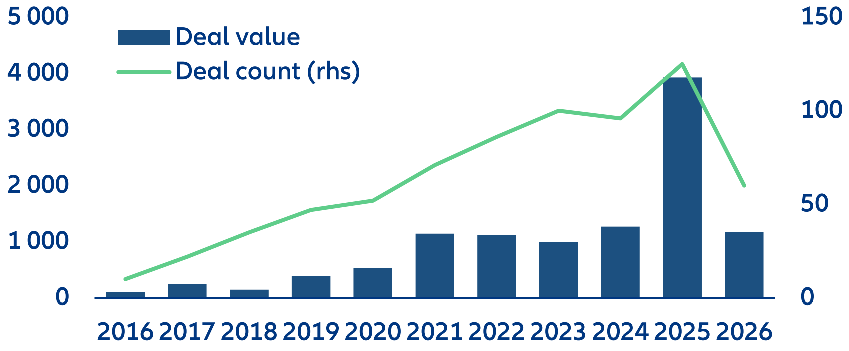
**The available loss estimates, though not comparable with each other, point in the same direction.** The BIS puts a restrained probability-weighted figure on the danger: financial-system losses equivalent to roughly 0.1% of GDP within five years of a quantum-enabled attack, rising above 1% over a 15- to 20-year horizon, with potentially larger effects in emerging markets. The Hudson Institute examines a more severe tail scenario, modelling a quantum-enabled cyberattack on the Fedwire interbank payment system using a two-stage approach: first calibrating the payment-network contagion effects of an attack on a systemically important bank (drawing on Federal Reserve researchers' Fedwire impairment simulations), then feeding that shock into Oxford Economics' Global Economic Model to trace the macroeconomic path. Their central estimate is a 10–17% decline in US real GDP, equivalent to USD 2.0–3.3trn in GDP at risk (3% of Word GDP). Hudson separately offers a cruder, earlier estimate — USD 726bn–1.95trn — for a single-day disruption to one of the five largest US banks' Fedwire access, extrapolated from prior work on quantum threats to the power grid and cryptocurrencies rather than from the full macro model; Hudson itself flags this figure as preliminary and imprecise. The assumptions and methodologies differ substantially across and even within these studies, so the numbers should not be read as forecasts or combined into a single estimate. Their common message is that a failure of financial cryptography would be a macro event with sizeable spillover in the economy, not an IT incident.

## Follow the money: capability is accelerating faster than readiness

**If the systemic stakes are this high, the natural question is whether investment is flowing toward the machines, the defenses or both.** The two are not equally visible: capability spending shows up cleanly in venture and public market data, while defending from the threat is happening behind closed doors. It concerns financial institutions' internal IT budgets, cybersecurity vendors and infrastructure-replacement programs, rather than in flows classified as quantum investment. So, the answer the data can give is necessarily partial and biased. Capital is flowing rapidly into quantum capability: Quantum-technology start-ups raised an estimated USD12.6bn in 2025, with more than

90% directed toward computing rather than sensing or communications. Within that total, venture investors deployed roughly USD3.9bn across around 125 rounds, more than triple the previous year (Figure 4).

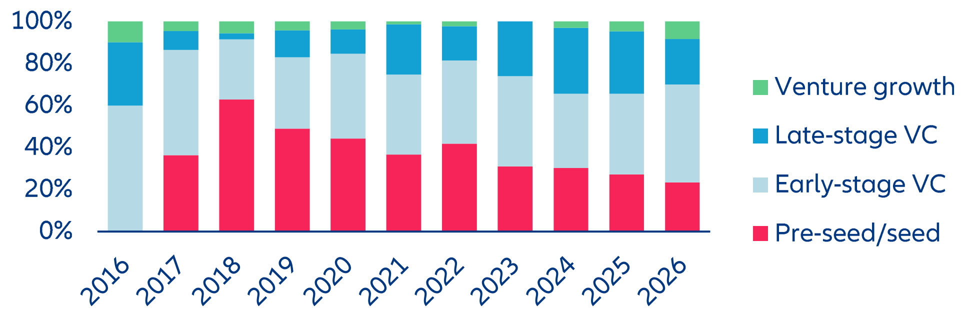
Figure 4: Quantum-computing VC deal activity by quarter, deal value (USD mn) and deal count




*Sources: PitchBook, Allianz Research. Note: 2026 is YTD values.*

**The composition of that capital has changed even faster than its size.** As recently as 2024, late-stage and venture-growth rounds were a rounding error; in 2025, venture growth alone jumped from around 1% to more than 30% of deal value, the largest single-year stage shift in the dataset, while mega-rounds of USD25mn and above quadrupled to roughly USD3.5bn. The investors writing those checks are no longer specialist quantum funds taking science risk but crossover institutions, strategic corporate balance sheets and sovereign wealth, some of the same names that anchor late-stage AI and semiconductor rounds. In capital markets terms, quantum has graduated from venture curiosity to institutional allocation. (Figure 5).

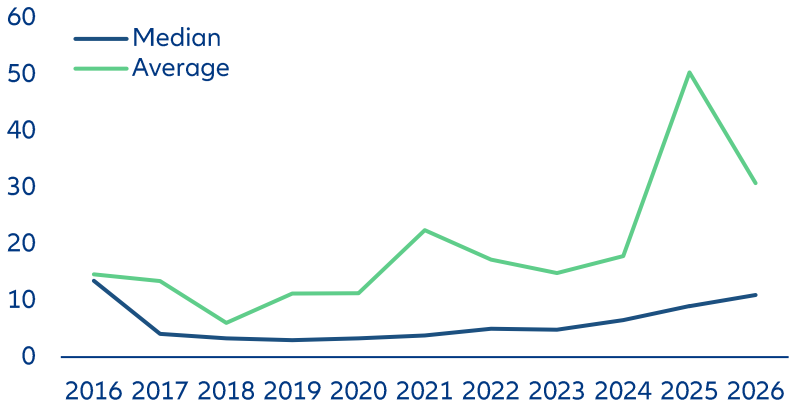
Figure 5 Share of quantum-computing VC deal value by stage




*Sources: PitchBook, Allianz Research*

**The surge in capital has not made the market any broader.** The category is governed by an unusually steep power law: the five largest deals account for roughly 27% of all venture value, and the five largest investors for more than 70%. The two largest rounds of the year were struck at pre-money valuations of roughly USD10bn and USD6bn, against a 2025 median pre-money valuation of under USD35mn. That gulf between the average and the median is the signature of a winner-take-most market, and it carries a direct portfolio consequence: the exposure is far easier to obtain than to underwrite. The listed proxies make the same point: pure-play quantum equities trade with betas of 2.5–3x the broad market, and the leading thematic ETF saw assets swell from around USD1bn to nearly USD6bn in 18 months. This is a beta bet on a narrative, not yet a cash-flow story (Figure 6).

Figure 6: Median and average quantum computing VC deal value, USD mn

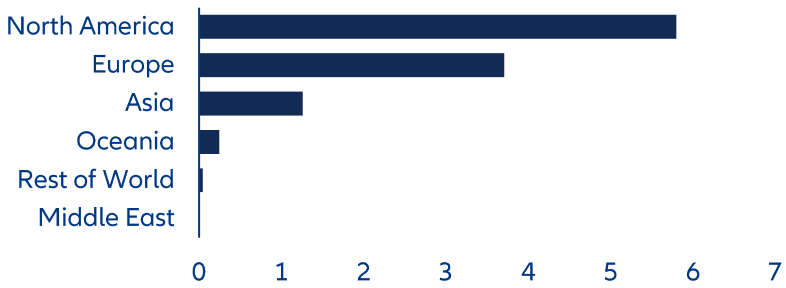



*Sources: PitchBook, Allianz Research*

**The harder truth sits on the other side of the holding period.** Quantum is deep-technology investing in its purest form, because it takes a decade or more from laboratory to liquidity and that horizon has so far left little room to exit: public-market debuts have generally underwhelmed, the underlying businesses are too young and too dependent on talent and intellectual property to suit a leveraged buyout and realized proceeds were close to nil as recently as 2024. The partial reopening in early 2026, a single quarter that produced more exit value than the previous three combined, reads more like a release of pent-up pressure than the arrival of a functioning exit market. The real conclusion is about duration rather than returns: the reward clock asks investors for a ten- to fifteen-year lockup, while the threat clock may strike inside a decade.

**The narrowness is geographic as well as financial.** Since the category took shape, roughly five of every six venture dollars have been committed in North America and Western Europe, while much of Asia's effort runs through state programs that never surface in venture statistics. Even within the West the split is lopsided: Europe attracts only around 5% of global private quantum investment, against more than half for the US, which is positioned to convert its advantages in capital markets, semiconductors and cloud infrastructure into technological leadership. The companies raising the money are also older than they used to be, with the median age at financing nearly doubling over the past decade from around two to four years. The picture is of a market that is deep but not wide: a concentrated pool of large, late and geographically clustered bets whose returns will be decided by a handful of outcomes rather than by the average (Figure 7).

Figure 7: Quantum-computing VC deal value by region, 2016-2026 (USD Bn)

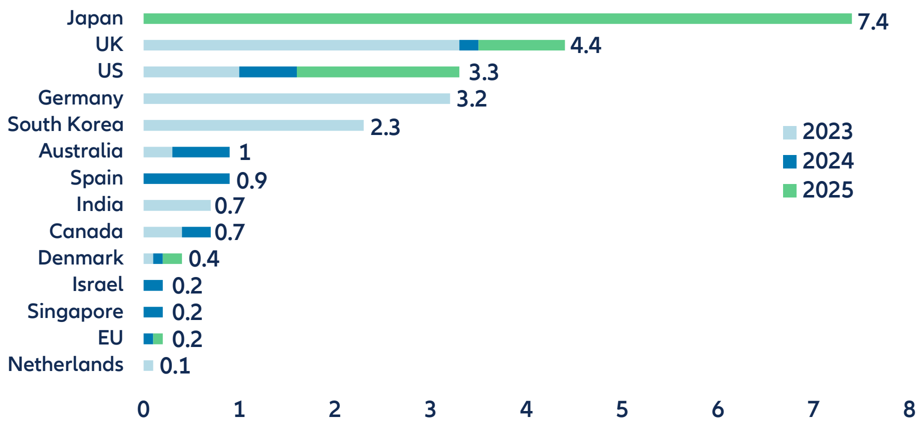



*Sources: PitchBook, Allianz Research*

**Most revealing of all is where the money goes.** Of every venture dollar committed to quantum, around 97% has gone to the technology stack – the qubits, the control systems, the platform layer – and less than 1% to financial-services applications. Private capital, in other words, is funding the machine's capability, not the applications that would benefit finance directly. The investor base financing that capability is, to a large extent, the same base that ultimately relies on the cryptographic foundations of the financial system, so the industry has a clear and shared interest in seeing resilience advance in step with capability. For now, investment in building the technology is running well ahead of investment in preparing for its consequences.

**Governments remain important sponsors, and in Washington, policy now addresses both clocks at once.** Around USD11.4bn of new public quantum-funding commitments were announced in 2025, led by Japan, the US, the UK and Spain, while earlier multi-year programs in Germany, South Korea and China remain substantial (Figure 8). The US executive order *Ushering in the Next Frontier of Quantum Innovation* seeks to accelerate computing, sensing and networking; strengthen supply chains and workforce development; expand public-private partnerships and deploy a scientific quantum computer at a Department of Energy facility. A parallel order, *Securing the Nation Against Advanced Cryptographic Attacks*, directs agencies to appoint migration leads, inventory cryptographic assets, pilot post-quantum systems and move high-impact federal systems toward post-quantum key establishment and signatures around the turn of the decade. The US is thus addressing capability and defense through a single, coordinated federal instrument.

Figure 8: Government announced commitments on quantum by year, USDbn




*Sources: McKinsey, Allianz Research. Note: Figures reflect press-search coverage of announced government commitments only, Jan 2023– Dec 2025; France's national quantum strategy (around €1.8B, Plan Quantique) falls outside this window as it was announced in 2021, and additional country-level commitments may be missing where they run through EU-level programs (e.g., Quantum Flagship) or lacked sufficient press coverage/breakdown to meet inclusion criteria. Figures should be read as directional signals of announcement activity, not a comprehensive accounting of national quantum investment.*

**Europe is not standing still, but its response is structurally more diffuse.** The European Commission's Quantum Europe Strategy, published in July 2025, targets European leadership by 2030, and a proposed Quantum Act, expected in 2026, would try to turn that ambition into industrial policy and coordinated funding. On the defensive clock, at least, Europe has set dates: The EU's coordinated post-quantum roadmap calls on member states to adopt national transition strategies by end-2026 and for critical infrastructure, including finance, to complete migration by 2030. On the capability side, however, the difficulty is fragmentation, which the strategy itself identifies as the central obstacle: European support is spread across Horizon Europe, EuroHPC, the Chips Act, national strategies and a dozen other instruments, without a comparably deep single market for scale-up capital. Where Washington can act through parallel executive orders, Brussels must first coordinate member states and programs. After artificial intelligence, quantum technologies risk becoming another arena in which the US converts a unified market and deeper capital into a durable lead, while Europe legislates and regulates a technology it does not yet own.

**The larger gap, however, is not between regions but inside institutions**. Only around 35% of large organizations have completed a comprehensive inventory of cryptographic assets, and a smaller share combines executive governance, budgets and a credible migration roadmap. Post-quantum migration is not principally a quantum-research project. It is a large IT and engineering program involving inventory, procurement, architecture, testing, certificate management, hardware replacement and coordination with vendors and counterparties. And the gap runs on both sides of the ledger: just as few institutions have a credible migration roadmap, few have moved beyond scattered pilots to the systematic talent, benchmarks and integration capacity needed to capture the reward when it arrives. Prepared institutions are the exception on either clock.

**Its binding constraints are therefore budget, project capacity and executive attention, the same resources being absorbed by artificial intelligence.** The competition is concrete and largely fought within the same IT budget line: the engineers, cloud spend and board time that could fund a cryptographic-migration program are the ones being directed toward AI, which offers more visible and immediate returns. Post-quantum defense risks losing that internal contest not because the threat is unknown, but because its payoff is an avoided loss rather than new revenue, and its deadline remains uncertain.

**None of this weakens the opportunity.** The investment data show genuine conviction, growing institutional participation and the early development of an industrial ecosystem, and the Monte Carlo case gives insurers in particular a legitimate reason to experiment on both sides of the balance sheet. But the discipline applied to capturing the upside should be matched by equal discipline on timing and resilience, because the two clocks reward different behavior. Capturing the reward is a wager on when quantum beats a moving classical benchmark, and it can afford to wait for evidence. Managing the threat cannot: Harvest-now-decrypt-later means today's traffic is already exposed to tomorrow's machine, and migration itself takes years. That is why quantum preparedness is best understood as an IT-resourcing decision rather than a bet on an arrival date: Crypto-agile architectures, prioritized inventories, migration roadmaps and tested fallbacks pay off across almost every timeline, while waiting pays off only in the one where the machine never comes. What to watch from here: a credible demonstration of factoring at cryptographic scale, logical-qubit counts scaling past error-correction thresholds, the first mandated live post-quantum payment rail, credible business advantages to the financial sector, enforcement of the 2030 deprecation deadlines and the annual shift in the expert threat timeline. Institutions do not need to know which clock will strike first to know that preparing for only one is the wrong strategy.

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(v) persistency levels, (vi) particularly in the banking business, the extent of credit defaults, (vii) interest rate levels, (viii) currency exchange rates including the EUR/USD exchange rate, (ix) changes in laws and regulations, including tax regulations, (x) the impact of acquisitions, including related integration issues, and reorganization measures,
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Allianz Research | 29 July 2026

# Diversification: Why bonds no longer hedge equities

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## In Summary

- **The 60/40 portfolio has quietly stopped working.** The risk-adjusted return investors could reasonably expect from the standard balanced portfolio has fallen to roughly a third of its long-run level, from a Sharpe ratio of 0.47 to just 0.14, as bonds and equities have stopped hedging each other for the first time in a generation. Since mid-2022, the two have moved together rather than apart, with the equity-bond correlation flipping to a positive +0.77, and no comparable episode in the past quarter-century has behaved this way for anywhere near this long.
- **The culprit is policy uncertainty, not only inflation.** The reassuring story that 2022 was a one-off inflation shock and that everything returns to normal once prices settle does not survive contact with the data. Inflation still matters, but it is no longer the master variable. Tariff policy, geopolitical fragmentation and the slow erosion of central bank credibility are now doing the heavy lifting.
- **Gold, private credit and infrastructure are the diversifiers doing the work now.** Where government bonds have failed to catch equity drawdowns since 2022, three asset classes have quietly stepped into their place. Gold has delivered positive risk-adjusted returns in nearly every regime tested and is the closest thing to an all-weather hedge. Private credit is posting its strongest showing on record. Infrastructure has held up across both this cycle and calmer ones before it, a rare cross-regime performer.
- **A regime-robust 'No-Regret' portfolio, built to work across cycles rather than optimized for the current one, roughly doubles the risk-adjusted return of a standard global benchmark.** It delivers a Sharpe ratio of 1.33 against the benchmark's 0.59, compounds an initial dollar to roughly 9x its starting value over two decades against just 5x for the benchmark and cuts the deepest drawdown in the sample from −24.6% to −15.3%. The composition marks a substantial departure from tradition: government bonds are cut from roughly 20% of the benchmark to under 4% and public equities from 43% to 13%, funding materially larger allocations to gold (16%), private credit (29%) and private equity (16%), with a smaller residual position in infrastructure.
- **The historical hedge will return only once policy uncertainty resolves and inflation is fully re-anchored, and rate cuts alone will not do it.** Only two of the five preconditions for a durable return to negative equity-bond correlation are true binding constraints, and neither is something a central bank can deliver on its own. Genuine resolution likely requires a durable Iran settlement and a more stable US trade regime, and both look like multiyear propositions rather than near term fixes. The remaining conditions, on monetary and liquidity settings, equity valuations and the yield curve, follow mechanically once the first two are in place.

## How 2022 snapped the 60/40 strategy

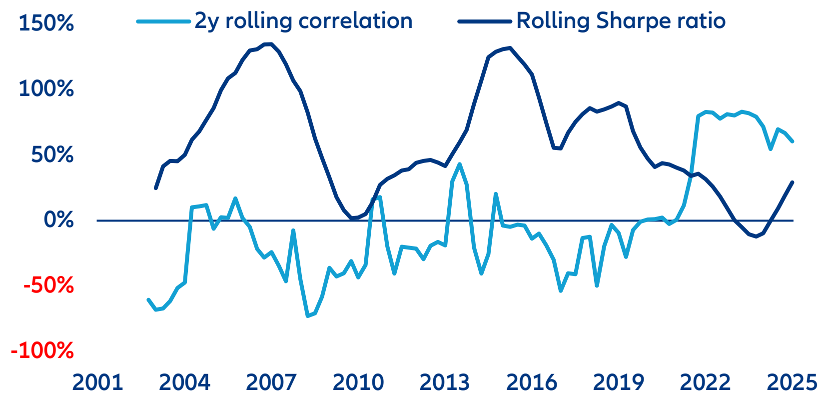
**For two decades, the 60/40 portfolio did something rather remarkable: it made the difficult look dull.** The formula was straightforward. Sixty cents of every dollar bought equities, which supplied the long-run growth. The other 40 went into government bonds, which stood by, patiently, waiting to catch the fall. When equities dropped, as they periodically did, bonds duly rose. Growth scares brought rate cuts, rate cuts pushed bond prices up and the 40% sleeve absorbed a decent share of the equity pain. Investors could speak, without embarrassment, of a portfolio that returned around 7% a year with the volatility of a mid-cap stock and the drama of an accountancy conference.

**But it was a convention, not a law of nature.** Yet it held often enough, and for long enough, that the entire architecture of modern retirement finance came to be built on top of it. Liability-matched pension books assumed it. Target-date funds mechanized it. An industry of insurance companies, endowments and family offices arranged themselves around the same quiet promise, that bonds would zig when equities zagged.

**Then, in 2022, the convention snapped and the wound is still open.** Between the early 2000s and 2021, a 60/40 book delivered a Sharpe ratio of roughly 0.47, a measure of how much excess return a portfolio earns per unit of risk taken (above 0.5 is respectable, above 1.0 enviable). It was unglamorous, but the sort of number pension trustees could take to their boards without flinching. In 2022, the hedge failed at the moment it was most needed. Equities and bonds sold off in unison, and the balanced portfolio delivered its deepest drawdown in a generation. A manager reviewing the year's returns could be forgiven for a moment of vertigo: the diversification premium, dependable for so long that it had ceased to feel like a premium at all, had simply vanished.

**Four years on, markets have not forgotten.** The Sharpe ratio implied by today's pricing on a 60/40 book sits closer to 0.14, less than a third of its long-run average. Portfolios that had been sold, quite reasonably, as boring have been forced to justify themselves anew (Figure 1).

Figure 1: When bonds stopped catching the fall (%)




*Sources: LSEG Datastream, Allianz Research*

**At this point the industry has, predictably, split into two camps.** The optimists take a relatively benign view. 2022 was an inflation shock and inflation shocks have always played havoc with the correlation between bonds and equities. Once inflation settles, and provided the world manages to avoid a second, 1970s-style episode, the old negative correlation should quietly reassert itself, as ordinary growth shocks reclaim the driver's seat. On this reading, patience is the only strategy required. The pessimists point to something rather harder to unwind. Fiscal and monetary policy have become, if not fused, then uncomfortably entangled. Geopolitical fragmentation is running at levels not seen since the end of the cold war. And central bank credibility, that intangible asset which underwrote so much of the previous cycle, has been slowly, publicly, corroded. On this view, the positive correlation can persist long after inflation itself has been tamed, because what has changed is not the price level but the framework around it.

**In this paper we apply both theories to post-2022 data and find that the pessimists have the better of it, though with an important wrinkle.** The dominant force behind today's positive correlation is not the monthly inflation print, awaited each month with such nervous attention on Wall Street, but policy uncertainty itself. Inflation still matters, of course. It shows up in the regression, alongside tighter liquidity, a flatter yield curve and stretched equity valuations, as a genuine driver of correlation. It is simply not the master variable. The implication is uncomfortable, particularly for those who have spent the past two years waiting for a single, clean catalyst. A first rate cut, however loudly cheered, will not on its own restore the hedge. Policy uncertainty is the binding constraint, and the bar for its removal is a good deal higher than markets are currently pricing. Investors counting on a return to the old regime may find themselves counting for some time.

## Where are we in the cycle?

**Five regimes, not one endless cycle.** The temptation, when confronted with the current mess, is to reach for the nearest historical analogue and declare it a rerun of the 1970s. This is almost always a mistake. Instead of assuming the past 25 years amount to a single continuous cycle, we sifted the macro data into five distinct regimes, each with its own equity-bond correlation and, if one is honest, its own personality. The first, which we call liquidity and rates normalization, ran from 2001 to 2008 and produced the workhorse negative correlation of -0.38, not spectacular, but exactly the diversification investors had learned to expect. (Correlation runs from minus one, a perfect mirror image, through zero, no relationship at all, to plus one, perfect lockstep; the more negative the number, the more reliably bonds cushion equity losses.) Crisis and deleveraging, the label for the acute shocks of 2008-09 and 2020, was brief but violent, and pushed the correlation to -0.69, the strongest hedge in the entire sample; when the world briefly ended, in other words, bonds worked harder than ever. Post-crisis reflation, the recovery years of 2009 to 2012, offered a more modest -0.19, with wide credit spreads and a steep yield curve. Then came the long risk on expansion of 2012 to 2022, a decade in which valuations rose, volatility slept and correlations grew lazy, flattening the hedge to -0.12.

**And then, in May 2022, the arrangement changed.** Today's regime is the odd one out. Elevated rates, stubborn inflation, high policy uncertainty and a compressed yield curve have combined to push the equity-bond correlation to a +0.77, a figure without precedent anywhere in the past 25 years.

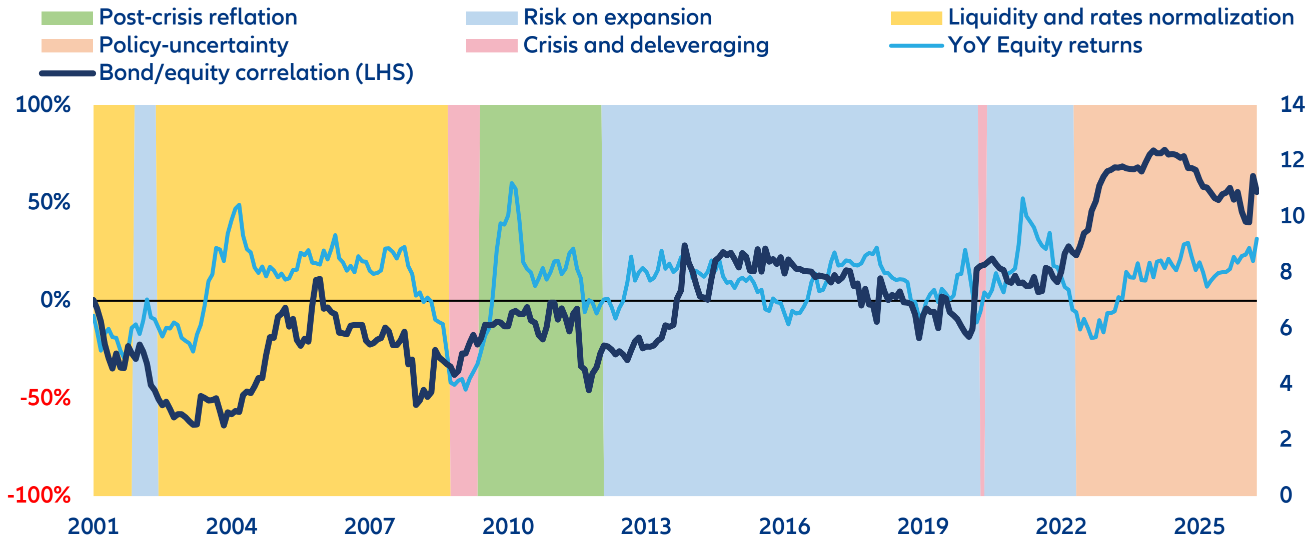
**Overlay the five regimes onto realized equity returns and the rolling bond-equity correlation and a familiar picture emerges: The hedge holds through every "normal" regime and breaks only when inflation becomes the dominant shock.** Through the dot-com bust and the subsequent recovery of 2003 to 2007, the correlation stays firmly negative, touching -60% even as equity returns swing from -40% to +40%. Rates and the shape of the curve did most of the work. The Fed entered 2001 with the funds rate still at 6.5%, cut aggressively through the dot-com bust to 1.75% by year-end and, because short rates fell faster than long ones, watched the curve steepen sharply even as growth expectations collapsed. That combination, elevated starting rates giving the Fed room to cut and a yield curve that could steepen on cue, is precisely the mechanism that lets bonds rally when equities sell off. It is, in other words, a textbook illustration of why bonds do their job as shock absorber at exactly the moment equities need them to.

**Then came 2008. The global financial crisis was, above all, a credit and volatility episode.** Those two forces dominated all other drivers, sentiment and growth expectations collapsed to sample lows and the recovery of 2009 and 2010 was a sharp, V-shaped affair driven by quantitative easing, which briefly, and unusually, pushed both equities and correlation higher together. What followed was the decade-long stabilization regime of 2011 to 2020, in which correlation settled into a shallow, range-bound negative-to-neutral band and equity returns unfolded in the calmer, low-rate manner that came to define the era. The Covid-19 shock of 2020 was narrow and sharp, marked by the highest readings in the sample for monetary and liquidity conditions as central banks intervened at unprecedented scale. It, too, resolved quickly.

**The regime that breaks this pattern is defined by policy, not growth, and that is precisely why we do not expect it to resolve on its own.** Since 2022, the correlation between bonds and equities has decoupled from its historical negative range and climbed into the 60% to 80% band, even as equity returns themselves have been comparatively muted. Unlike 2001, the Fed entered this shock with rates near zero, and there was, therefore, no room to cut when inflation and growth diverged. Instead, consumer price inflation and commodity inflation reached their highest levels in the sample, the dollar strengthened, equity valuations sold off, and hiking rather than cutting pushed bonds and equities in the same direction.

**By 2024, the character of the shock had rotated. Inflation itself cooled, but policy uncertainty and interest rate levels took over as the elevated forces, while monetary and liquidity conditions sank to sample lows.** Tariff policy and geopolitical shocks now sustain the same positive-correlation dynamic that inflation started. This, more than anything, is why the current regime should be treated as structurally different, not merely a noisier version of the stabilization years that preceded it (Figure 2).

Figure 2: A quarter-century in five acts (%)




*Sources: LSEG Datastream, Allianz Research; Note: 2y correlation*

**The current regime is structurally different, not simply unusually long.** Three features set it apart from every prior episode. The first is duration and consistency, every single quarter since May 2022 has fallen into this regime, and the average correlation has stayed stubbornly above +0.40 throughout. There is no precedent in the sample for that kind of persistence. The second is the identity of the dominant signal. It is not merely inflation, as many assumed at the outset, but policy uncertainty specifically. Our Economic Policy Uncertainty index, expressed as a z-score (that is, in standard deviations away from its long-run average, so that +1 means one standard deviation above trend and anything above +2 is unusually high), sits at +1.7 as of May 2026. It has been elevated since 2024, and briefly touched a remarkable +5.4 on "Liberation Day". The third, most awkward of the three is that this regime has never ended. Every quarter since May 2022 has remained inside it, with no observed exit anywhere in the historical record.

**There are two plausible exits from here, and one rather less likely one.** A transition into the risk on expansion regime, in which policy uncertainty resolves and liquidity conditions ease, is one credible route; a transition into the liquidity-normalization regime, in which rates settle into a predictable path and broad liquidity returns to the system, is the other. Both are structurally consistent with the way prior regimes have unwound. A direct jump into an acute crisis regime is also possible, though far less probable, and historically brief. Those episodes have, on average, lasted only around five months before resolving into something calmer.

## What is driving the change in correlation? An anatomy of a broken hedge

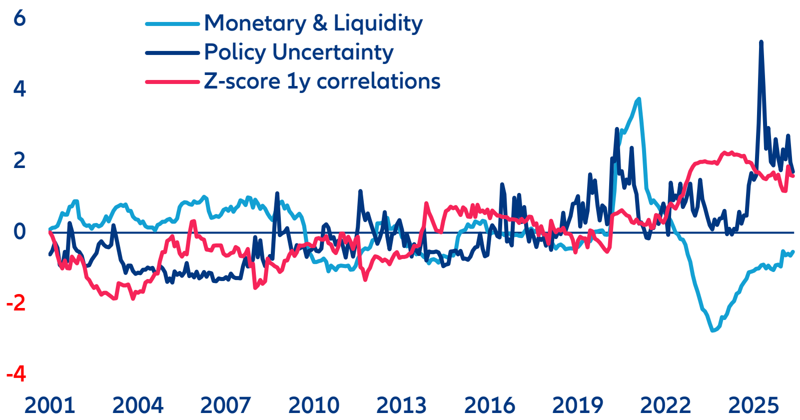
**Policy uncertainty is the biggest driver, but inflation matters too.** To work out which macro forces have actually been moving the equity-bond correlation, we regressed 25 years of rolling correlation data against five candidate factors. The results are unambiguous. Roughly two-thirds of the variation in correlation can be traced to those five drivers, and all five clear the usual statistical thresholds comfortably. Policy uncertainty emerges as the dominant force, pushing correlation firmly into positive territory whenever it rises. Monetary and liquidity conditions carry an almost identical, but opposite, effect, looser conditions restore the traditional hedge, tighter ones destroy it. Equity valuations turn out to matter more than most would guess. A stretched market pushes correlation up, a calmer one pulls it back down. A steeper yield curve, the mark of a "normal" rate environment, works quietly on the negative side. And inflation, contrary to those hoping the story is a pure policy-uncertainty morality play, also pushes correlation up in its own right, an effect that has been especially pronounced since 2022.

**That inflation still matters is a corrective to the pure policy uncertainty story.** The two forces have moved together, often, over the past two decades. In 2008 and 2020, the monetary and liquidity indicator surged to nearly 3.8 standard deviations above its long-run average, and correlation briefly flipped positive alongside extreme easing. In 2021, correlation turned positive ahead of the inflation shock itself. The two then rose together as consumer prices climbed, with CPI reaching 3.9 standard deviations above trend in 2022 while correlation's z-score ran up to a peak near 2. The post-2021 regime, in other words, was inflation-driven before it became uncertainty-driven.

**The character of the shock rotated after that.** In 2022, monetary and liquidity conditions collapsed to almost 2.8 standard deviations below trend as the Fed hiked, yet correlation stayed elevated: growth was no longer the master shock, inflation had taken over. Inflation then receded faster than correlation, which anchored near 1.5 to 2 through 2024 and 2025 as policy uncertainty stepped in as the marginal driver, consistent with the tariff-driven uncertainty of the second Trump administration. The sharpest reading in the sample, policy uncertainty touching 5.4 in early 2025, captures the “Liberation Day” shock. Uncertainty then remained stubbornly high through the escalation to "major combat operations" in the February 2026 Iran conflict, a longer and more consequential episode than the brief Israel-Iran strikes of June 2025 (which had only pushed Brent crude from around USD65 to the low USD80s before subsiding). All of it feeds the same inflation-and-uncertainty channel that has kept bonds and equities correlated near their sample highs.

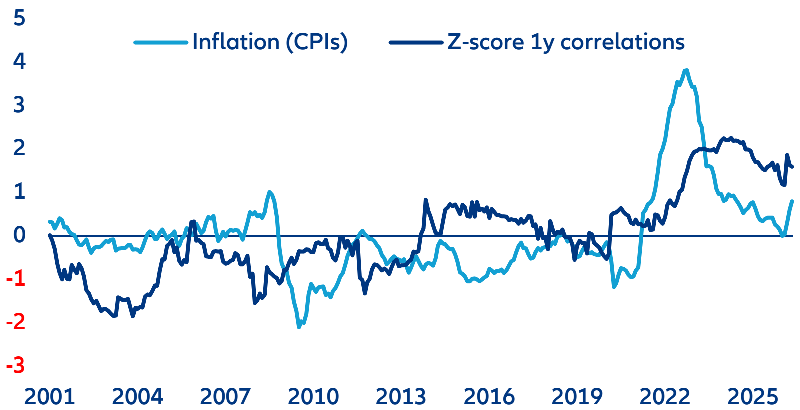
**The model, reassuringly, still works.** Its predictions match reality most closely when one or two forces dominate the regime, and most poorly when short, sharp shocks temporarily overwhelm the underlying macro drivers. The fit is at its best during the 2001 to 2008 and 2022 to 2026 windows, precisely when a small number of factors have been doing most of the driving, rates and the yield curve in the early 2000s, and policy uncertainty and monetary policy together from 2022 onwards. It tracks worst around the 2009 to 2010 rebound and the 2020 Covid-19 dip, when actual correlation overshot the model in both directions faster than macro variables could move. Those, unsurprisingly, are precisely the moments when positioning, liquidity dislocations and forced deleveraging drive realized co-movement independent of the fundamentals the model measures. The gap widens for a while, then both series converge again. Feeding May 2026 inputs into the model produces a predicted correlation reading of 1.05 standard deviations above the long-run average, against an actual reading of 1.59. Policy uncertainty is the largest contributor at +0.62, with equity valuations, monetary and liquidity conditions and inflation each contributing between +0.11 and +0.14, and a re-steepening yield curve chipping in a marginal +0.04. The gap of +0.54 between prediction and reality is larger than in prior months, consistent with a degree of regime persistence the underlying factors alone do not capture. The correlation has become, in effect, self-reinforcing. Portfolio managers have already cut their duration exposure in response to it, and their reduced participation keeps realized co-movement elevated even as the underlying drivers ease a fraction (Figure 3 and 4).

Figure 3: Where the correlation comes from (Z-scores)




*Sources: LSEG Datastream, Allianz Research*

Figure 4: The model, put to the test (Z-scores)



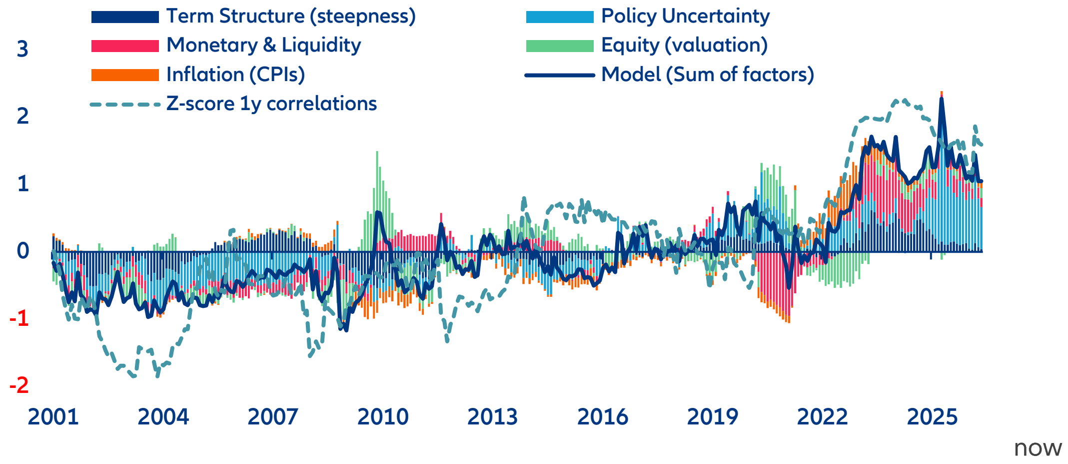

*Sources: LSEG Datastream, Allianz Research*

**What has to happen before the hedge normalizes? A durable return to negative bond-equity correlation requires five conditions to line up.** The good news is that only two of them are genuine constraints. The other three are largely endogenous to a normal easing cycle and should follow, mechanically, once it is underway.

**The hard ones first. Policy uncertainty must recede, and inflation must remain on a credible disinflation path.** Neither is something a central bank can deliver on its own, which is why we do not expect the hedge to return simply because the Fed or the ECB begin cutting. Policy uncertainty today is anchored by an unresolved geopolitical and trade backdrop, with the Iran conflict's ceasefire collapsing again in July 2026 and tariff policy still being litigated through the courts and alternative executive authorities. Genuine resolution likely requires a durable Iran settlement and a more stable US trade regime, and both look like multi-year propositions rather than near-term fixes. Inflation, for its part, has retraced most of its 2022 surge, but tariff pass-through and Gulf-driven energy volatility keep the disinflation path only partially secured.

**The remaining three conditions should follow of their own accord.** Monetary and liquidity conditions will loosen as the new Fed leadership under Kevin Warsh, who was expected at his May 2026 installation to lean toward easier policy but has so far held a more neutral stance, continues to cut rates. Equity valuations, a channel of similar size to policy uncertainty and liquidity in the updated regression, are likely to moderate as the AI capital-spending cycle cools or broadens into green-transition and infrastructure investment rather than through an outright shock. And the yield curve should steepen as a natural consequence of rates normalizing once the first four conditions are in train. Investors waiting on the first cut, in short, are waiting on the wrong thing (Figure 5).

Figure 5: The road back to a working hedge (Z-scores)



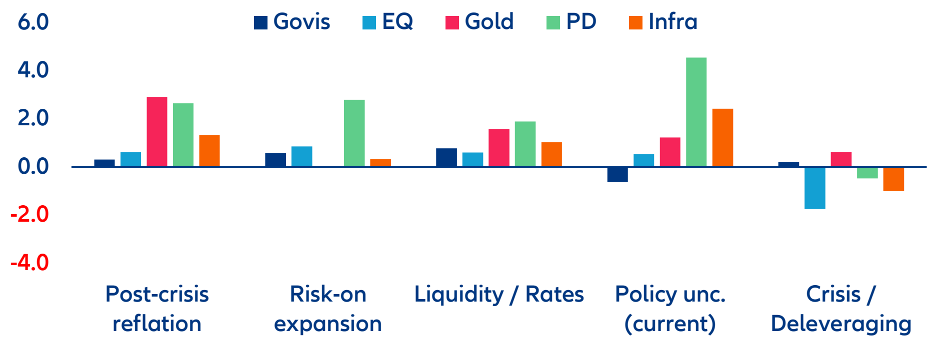

*Sources: LSEG Datastream, Allianz Research*

## What are the implications for portfolio construction?

**Some asset classes are earning their keep.** In the current regime, the 15 asset classes we track are being sorted rather ruthlessly. Government bonds, agencies, investment-grade credit and inflation-linked bonds have all posted negative risk-adjusted returns in Regime 4 (the current policy-uncertainty and inflation regime, running since May 2022), the direct consequence of losing the very negative correlation that once made them portfolio anchors. Government bonds alone have swung from a positive Sharpe ratio in every other regime (between +0.22 and +0.78) to -0.64 in the current one. Public equities have held up, more or less, but without the tailwind of a hedge behind them: a modest +0.54, well short of their strongest showing during the risk-on expansion of the 2010s (+0.86) but nowhere near the collapse they suffered in the crisis and deleveraging regime (-1.75).

**Three asset classes stand out as the genuine diversifiers of this cycle.** Gold has delivered a positive Sharpe ratio in nearly every regime we tested, including the current one at +1.22, and its finest hour came in the post-crisis reflation of 2009 to 2012, at +2.92. It is the closest thing to an all-weather hedge in the dataset. Private credit has produced its best risk-adjusted performance of any regime in this one, at +4.55, quietly stepping in as a source of income where traditional fixed income has been failing. Infrastructure, meanwhile, has been the standout real asset, reaching +2.43 in the current regime while also holding up in calmer, more traditional regimes such as liquidity and rates normalization (+1.03) and post-crisis reflation (+1.34). It is one of the very few exposures with strong performance across both the current regime and the more typical ones that came before: a genuinely cross-regime diversifier rather than a regime-specific bet.

Figure 6: Where risk pays and where it doesn't (Sharpe Ratio (rf=2%, annualized))




*Sources: LSEG Datastream, MSCI, Allianz Research*

**In the face of unexpected regime shifts, it makes sense to plan for the transition, not just the present.** Regime 4 has a perfect track record, in our sample, of persisting rather than resolving. The more useful planning question, then, is not whether it will eventually end but what should change once it does. The two structurally plausible exits both point in the same direction. As policy uncertainty fades and liquidity normalizes, government bonds and investment-grade credit should regain a meaningful share of their traditional hedging role, while the current outperformers – private credit, infrastructure and gold – would see their relative advantage compress back toward more typical historical levels. That argues for holding today's tilts with conviction, but not permanently.

**The No-Regret portfolio we define is built to perform across regimes, not just this one.** A brief word on how we got there. The optimization blends three ingredients in a way that deliberately avoids over-relying on any single one. It starts from a market-capitalization-style prior allocation, which anchors the portfolio to the actual composition of investable capital in the world. It then nudges those weights using the return signal implied by market pricing itself, a standard technique known as Black-Litterman, which avoids the notorious instability of feeding raw historical averages straight into an optimizer. And it adds a modest tilt from a machine-learning model trained on each asset's own momentum and drawdown history. Diversification, turnover and tail-risk penalties are built into the objective directly and the final result is stress-tested by resampling across thousands of simulated histories rather than optimized against a single historical path. The final weights, in short, are not simply the one combination that happened to work best in hindsight.

**The portfolio calculations are a mathematical exercise rather than a bespoke recommendation for any individual investor.** As discussed, the optimizer does embed a series of modeling choices we set directly rather than resulting from historical return data alone. However, it does not account for market depth, transaction costs, asset manager fees, taxes or an investor's specific liquidity needs or mandate constraints. We assume a risk-tolerant, unbiased and unrestricted investor purely relying on mathematical concepts, which is rarely the case in practice.

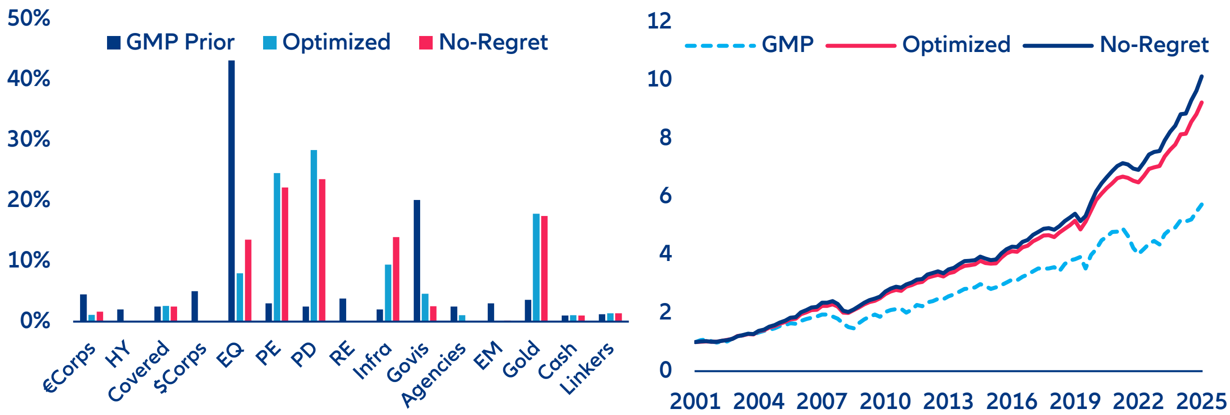
**As such, the output should be read as an analytical starting point rather than an end point**: a way to stress-test conventional allocations and surface where the mathematics disagrees with convention, with the actual translation into an investable portfolio left to an investment professional who can weigh it against real-world constraints.

**On a full-sample, risk-adjusted basis, the No-Regret portfolio outperforms both a traditional benchmark and a portfolio optimized purely for the current regime.** Optimizing a portfolio for the current regime works beautifully if the regime persists indefinitely, and rather less well the moment it does not. Instead, we constructed a blended No-Regret portfolio, using an equal-weighted average of all five regime correlations, with the current regime given double weight to reflect its higher-for-longer persistence. The blended equity-bond correlation embedded in the construction comes to +0.037, essentially neutral, against -0.032 using the full-sample quarterly correlation and +0.77 using the current regime alone. The result is a portfolio anchored around private markets and gold, with a reduced equity allocation and a small residual government bond position kept specifically as insurance for the eventual regime transition.

**The shift away from the Global Market Portfolio (GMP) prior, which is roughly the market-capitalization-weighted mix of all investable assets globally and serves as the optimizer's neutral starting point, is large, consistent across methodologies and has paid off for two decades, not just recently.** Private debt rises from a 2.5% prior weight to 28.5% in the No-Regret allocation, private equity from 3.0% to 15.9% and gold from 3.6% to 16.0%, while government bonds fall from 20.1% to just 3.6% and equities are cut from 43.2% to 12.6%. Reassuringly, an alternative construction built directly on the unconditional full-sample correlations rather than the regime blend lands in a broadly similar place on most positions. The tilt toward private markets and gold, in other words, does not appear to be an artefact of regime-weighting choices, it holds up even without leaning on the current regime at all.

**The payoff has been persistent, not a recent flourish.** Both the optimized and No-Regret portfolios compound from 1.0 to roughly 9x over the sample, clearly separating from the GMP benchmark's roughly 5x outcome by the mid-2010s and holding that gap through both the 2020 and 2022 drawdowns, consistent with Sharpe ratios of 1.35 and 1.33 against the benchmark's 0.59.

Figure 7: Two decades of compounding (allocation weights – left , cumulative NAV – right)




*Sources: LSEG Datastream, MSCI, Allianz Research*

**On that basis, the No-Regret portfolio posts the second-highest Sharpe ratio of any portfolio we tested (1.33),** behind only the full-sample optimized portfolio (1.35), which has the significant benefit of hindsight, and comfortably ahead of the Global Market Portfolio benchmark (0.59). Its maximum drawdown, at minus 15.3%, is meaningfully shallower than the benchmark's minus 24.6%, and notably better than a portfolio built purely around current-regime assumptions (minus 18.9%). That last portfolio, despite looking attractive on paper today, offers the weakest downside protection of the group once tested across the full cycle.

Table 1: Risk-adjusted performance across portfolio constructions

| Portfolio | Sharpe ratio | Annualized return | Max drawdown |
|---|---|---|---|
| Global Market Portfolio (GMP) | 0.59 | 7.1% | –24.6% |
| Optimized for current regime (R4) | 1.07 | 10.0% | –18.9% |
| Full-sample optimized | 1.35 | 9.61% | –12.9% |
| No-Regret portfolio | 1.33 | 9.9% | –15.3% |

*Sharpe ratio uses rf = 2%. Figures based on quarterly data, 2001–2025 (97 observations).*

**Relative to a typical diversified benchmark, the resulting No-Regret mix makes three deliberate trade-offs: less public equity, materially less government bond duration and a substantially larger allocation to gold and private markets, particularly private credit and infrastructure.** The optimizer switches public for private counterparts: Private equity offers higher returns with lower volatility than public equity; private debt carries slightly more volatility than government bonds but far higher returns. None of these tilts is extreme or all-in. They represent a rebalancing of emphasis toward the exposures that have actually protected portfolios through this cycle, without abandoning the assets that will matter again once the regime eventually turns.

**The No-Regret allocation looks private.** The optimizer overweight private markets and gold for a straightforward reason. These exposures combine higher risk-adjusted returns with genuinely low correlation to a standard 60/40 book, which is precisely the combination a mean-variance optimizer is built to chase. The diversification case is visible in the data. US private equity, private debt, and infrastructure move through distinctly different cycles rather than in lockstep: private equity's mid-cycle strength around 2016 to 2019 arrives as infrastructure is flat or negative, infrastructure's sharp 2016 and 2022 spikes come while private debt is subdued and the four series only really converge during the 2008 to 2009 and 2020 shocks, when almost everything sells off together.

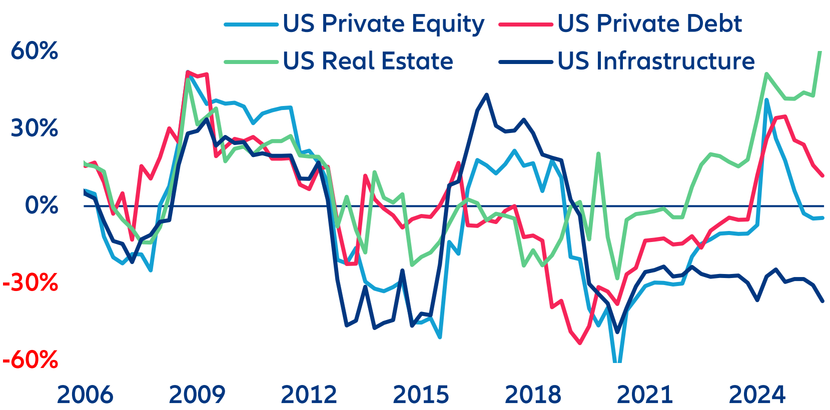
Table 2: Recommended allocation versus a typical diversified benchmark

| Asset class | GMP (typical mix) | No-Regret (recommended) | Difference |
|---|---|---|---|
| Public equities | 43.2% | 12.6% | –30.6pp |
| Private equity | 3.0% | 15.9% | +12.9pp |
| Private credit | 2.5% | 28.5% | +26.0pp |
| Infrastructure | 2.0% | 13.5% | +11.5pp |
| Government bonds | 20.1% | 3.6% | –16.5pp |
| Gold | 3.6% | 16% | +12.4pp |

*GMP reflects a market-capitalization-weighted global benchmark spanning public and private asset classes. Figures are illustrative allocations, not personalized investment advice.*

**That pattern of usually uncorrelated, occasionally-jointly-stressed returns is precisely what shows up in the regime-level Sharpe ratios.** Private debt posts its best risk-adjusted performance of any regime in the current one (+4.55). Infrastructure holds up across the current and calmer regimes alike (+2.43 today, and between +1.03 and +1.34 in earlier non-crisis eras). Gold, meanwhile, is positive in nearly every regime we tested. Higher Sharpe plus low correlation to the traditional stock-bond mix is, in a word, what the optimizer is rewarding when it reallocates away from equities and government bonds and into these exposures.

Figure 8: Private assets, dancing to their own beat (three-year rolling correlation of private assets with 60/40 portfolio)



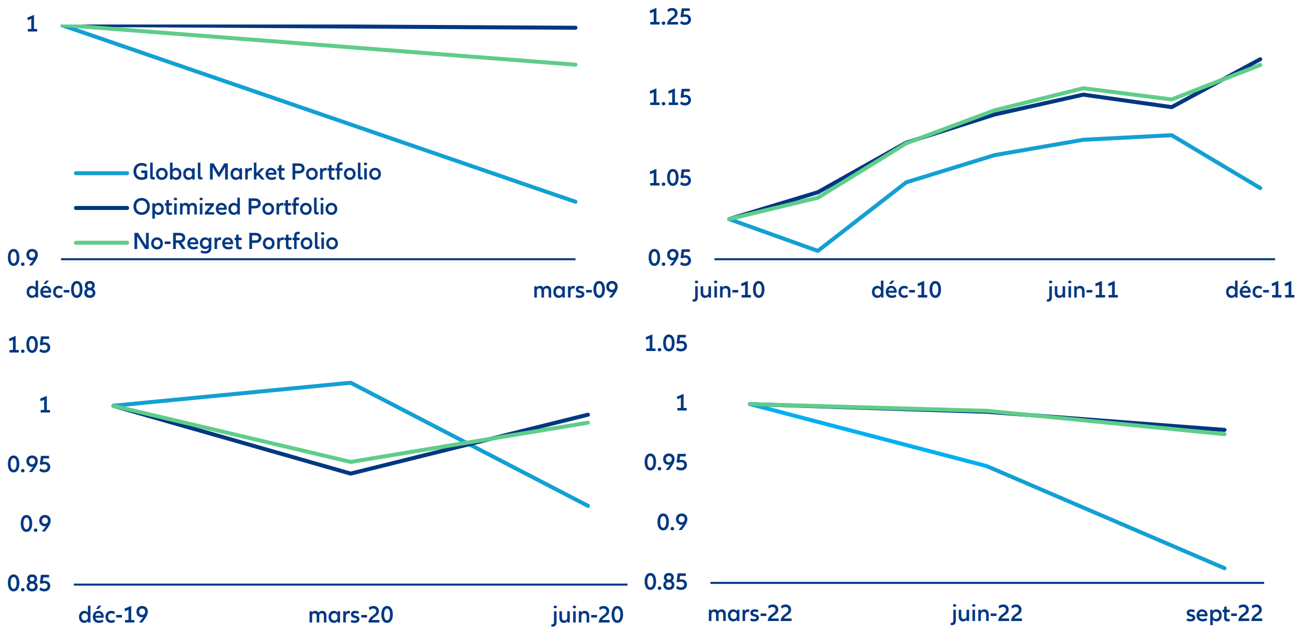

*Sources: MSCI, Allianz Research*

**One important caveat. The shift into private assets is only appropriate for investors with genuine capacity to bear illiquidity risk.** Private markets cannot be rebalanced or exited on short notice the way public equities and government bonds can. The allocation therefore assumes a long investment horizon, limited near-term liquidity needs and a governance structure able to tolerate multi-year lock-ups and unpredictable capital calls. Where that capacity is absent, the same diversification logic points to a smaller private-markets tilt, funded from public equity rather than from the liquid fixed income that may still be needed for near-term cash flow.

**The cost of choosing robustness over point-optimality is small and well worth paying.** The fully-optimized, full-sample portfolio edges out the No-Regret construction on both Sharpe ratio (1.35 versus 1.33) and drawdown (minus 12.9% versus minus 15.3%). But it does so only by implicitly assuming that the benign full-sample correlation structure persists, which is precisely the assumption this paper's diagnosis calls into question. Giving up 0.02 of Sharpe and roughly two additional points of drawdown, in exchange for a portfolio that is not quietly betting on the current regime never ending, is an easy trade.

**The crisis-period behavior makes the case for that insurance plain.** In the global financial crisis, the No-Regret portfolio held its NAV within two points of the fully-optimized one, while the Global Market Portfolio benchmark lost roughly 8%. In the 2022 rate shock, the one crisis actually built on the current regime itself, both No-Regret and the optimised construction held up far better than the benchmark, even as the portfolio built purely for today's regime actually gained ground. That last outcome is a useful reminder: the regime-specific optimum wins only if the regime in question is the one that actually materializes.

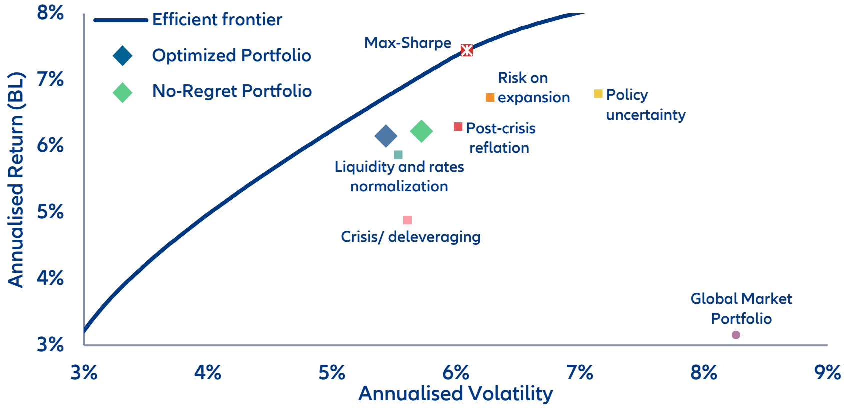
Figure 9: How each portfolio held up when it mattered (return rebased to 1)




*Sources: LSEG Datastream, MSCI, Allianz Research*

**The Monte Carlo frontier confirms this is not a free lunch dressed up as one.** No-Regret sits close to the efficient frontier with a Sharpe of 0.74, just behind the fully-optimized point (0.76) and ahead of every single-regime-optimal portfolio except Regime 2's risk-on construction (0.75), including the Regime 4-specific optimum (0.67 Sharpe, though R4 does carry the highest raw return among the regime portfolios, reflecting a bet on today's regime persisting). It sits well clear of the Global Market Portfolio benchmark (0.14). The frontier's true mathematical ceiling sits higher still, at a maximum Sharpe of 0.89, but only by concentrating nearly 70% of the portfolio in private debt alone, with gold, private equity and infrastructure making up almost all of the remainder and ten of the fifteen asset classes receiving no allocation at all. That portfolio is a mathematical artefact, not something any real mandate would hold. Taken together, the evidence says the No-Regret portfolio gives up only a modest amount of Sharpe (both against the fully-optimized point and against the true theoretical ceiling) in exchange for a portfolio that is actually diversified, and for materially better protection against downside risk and the one scenario this whole paper is built around: the current regime eventually ending.

Figure 10: Robustness, for a sliver of Sharpe




*Sharpe ratios on this chart use Black-Litterman-blended (BL) returns, a shrinkage-adjusted version of historical averages. This is why the same portfolios (e.g. GMP, No-Regret) show different Sharpe ratios here than in Table 1.*

*Sources: LSEG Datastream, MSCI, Allianz Research*

These assessments are, as always, subject to the disclaimer provided below.